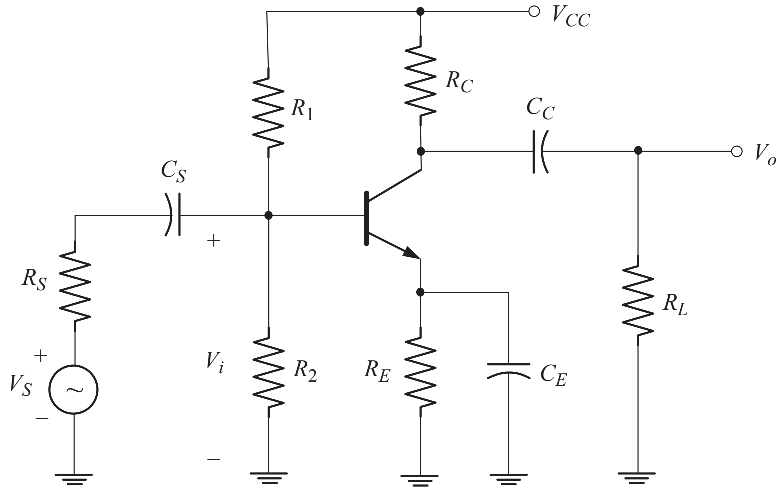


Fig. 4.13 Single stage BJT amplifier with coupling and bypass capacitors

Effect of Input Coupling Capacitor C_S on Low Frequency Response

The input coupling capacitor C_S couples the source signal to the active device (BJT). To study the effect of C_S on low frequency response we have to neglect the effects of C_C and C_E. This can be done by treating them as short circuits.

Now let us obtain the ac equivalent circuit by reducing V_{CC} to zero and replacing C_C and C_E by their short circuit equivalents. C_S is retained as it is. The ac equivalent circuit is shown in Fig. 4.14.

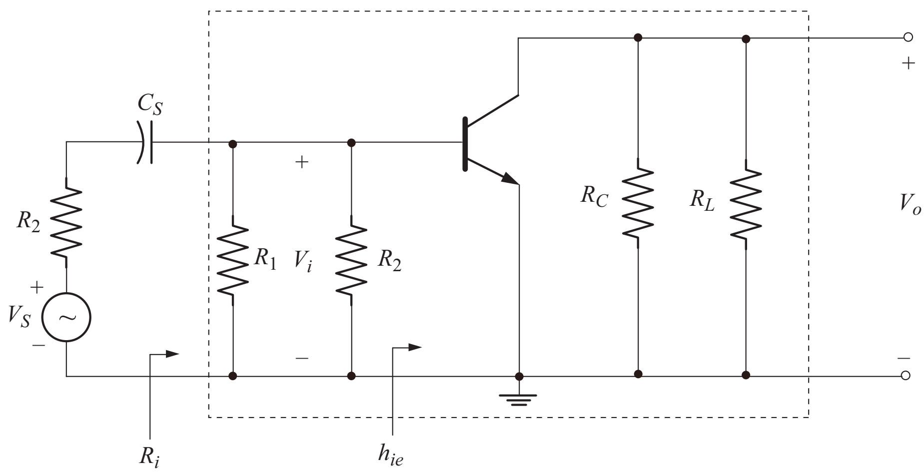


Fig. 4.14 AC equivalent circuit

The resistance of the transistor between base emitter is h_{ie}. The input ac equivalent circuit is shown in Fig. 4.15.

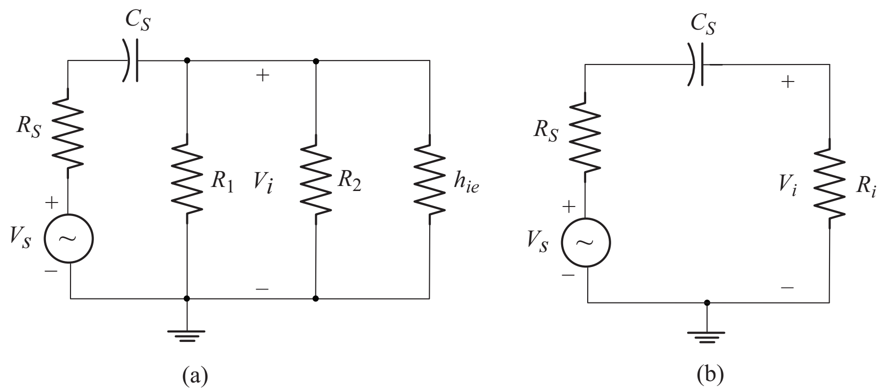


Fig. 4.15 (a) Input ac equivalent circuit
(b) Simplified input ac equivalent circuit

Let $$R_i = R_1 \| R_2 \| h_{ie} \tag{4.18}$$

where $$h_{ie} = \beta\, r_e \tag{4.19}$$

Using voltage division rule in the circuit of Fig. 4.15(b), the voltage applied to the amplifier is given by

$$V_i = \frac{V_S\, R_i}{[R_S + R_i] - j\, X_{C_S}} \tag{4.20}$$

where $$X_{C_S} = \frac{1}{2\pi\, f\, C_S} \tag{4.21}$$

$$V_i = \frac{V_S\left[\dfrac{R_i}{R_S + R_i}\right]}{1 - j\left[\dfrac{X_{C_S}}{R_S + R_i}\right]}$$

$$|\,V_i\,| = \frac{|\,V_S\,|\left[\dfrac{R_i}{R_S + R_i}\right]}{\sqrt{1 + \left[\dfrac{X_{C_S}}{R_S + R_i}\right]^2}} \tag{4.22}$$

In the mid frequency band, f is sufficiently large. As a result, $X_{C_S} \to 0$. Now from Equation (4.22) we have

$$|\,V_i\,|_{\text{mid}} = \frac{|\,V_S\,| R_i}{R_S + R_i} \tag{4.23}$$

Using this relation in Equation (4.22), we have

$$|V_i| = \frac{|V_i|_{\text{mid}}}{\sqrt{1+\left[\dfrac{X_{C_S}}{R_S+R_i}\right]^2}} \tag{4.24}$$

The lower 3 dB cut-off occurs, when

$$|V_i| = \frac{|V_i|_{\text{mid}}}{\sqrt{2}} = 0.707\,|V_i|_{\text{mid}}$$

From Equation (4.24) we find that, the 3 dB cut off occurs, when

$$\frac{X_{C_S}}{R_S+R_i} = 1 \quad \text{or} \quad X_{C_S} = R_S + R_i$$

$$\frac{1}{2\pi f\, C_S} = R_S + R_i$$

or
$$f = \frac{1}{2\pi\left[R_S+R_i\right]C_S} \tag{4.25}$$

Equation (4.25) gives the lower 3dB cut-off frequency due to C_S. Let us denote it by f_{L_S}. Now we have

$$f_{L_S} = \frac{1}{2\pi\left[R_S+R_i\right]C_S} \tag{4.26}$$

Effect of Output Coupling Capacitor C_C on Low Frequency Response

The output coupling capacitor C_C couples the output of the active device to the load. To study the effect of C_C on low frequency response, let us neglect the effect of C_S and C_E by treating them as short circuits.

From the circuit of Fig 4.14, we can write the ac equivalent circuit on the output side as shown in Fig. 4.16(a).

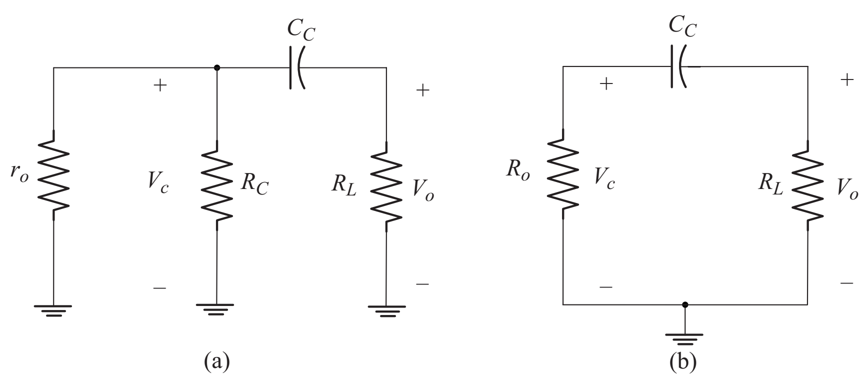


Fig. 4.16 (a) AC equivalent circuit of output side
(b) Simplified ac equivalent circuit

Let $$R_o = r_o \parallel R_C \tag{4.27}$$

The simplified ac equivalent circuit is shown in Fig. 4.16(b).

V_c = Output voltage of active device

V_o = Load voltage

Using voltage division rule in the circuit of Fig. 4.16(b), the load voltage is given by

$$V_o = \frac{V_c R_L}{[R_o + R_L] - jX_{C_C}} \tag{4.28}$$

where $$X_{C_C} = \frac{1}{2\pi f C_C} \tag{4.29}$$

$$V_o = \frac{V_c\left[\dfrac{R_L}{R_o + R_L}\right]}{1 - j\left[\dfrac{X_{C_C}}{R_o + R_L}\right]}$$

$$|V_o| = \frac{|V_c|\left[\dfrac{R_L}{R_o + R_L}\right]}{\sqrt{1 + \left[\dfrac{X_{C_C}}{R_o + R_L}\right]^2}} \tag{4.30}$$

In the mid frequency band, $X_{C_C} \to 0$. Now from Equation (4.30) we have

$$|V_o|_{\text{mid}} = \frac{|V_c|\, R_L}{R_o + R_L} \tag{4.31}$$

Using this relation in Equation (4.30) we have

$$|V_o| = \frac{|V_o|_{\text{mid}}}{\sqrt{1 + \left[\dfrac{X_{C_C}}{R_o + R_L}\right]^2}} \tag{4.32}$$

The lower 3dB cut-off occurs when

$$|V_o| = \frac{|V_o|_{\text{mid}}}{\sqrt{2}} = 0.707\, |V_o|_{\text{mid}}$$

From Equation (4.32) we find that, the 3dB cut-off occurs when

$$\frac{X_{C_C}}{R_o + R_L} = 1 \quad \text{or} \quad X_{C_C} = R_o + R_L$$

$$\frac{1}{2\pi f C_C} = R_o + R_L$$

or
$$f = \frac{1}{2\pi\left[R_o + R_L\right]C_C} \tag{4.33}$$

Equation (4.33) gives the lower 3 dB cut-off frequency due to C_C. Let us denote it by f_{L_C}. Now we have

$$f_{L_C} = \frac{1}{2\pi\left[R_o + R_L\right]C_C} \tag{4.34}$$

Effect of Emitter Bypass Capacitor C_E on Low Frequency Response

To study the effect of C_E on low frequency response, let us neglect the effect of C_S and C_C by treating them as short circuits.

From the circuit of Fig. 4.13, we can write the ac equivalent as shown in Fig. 4.17.

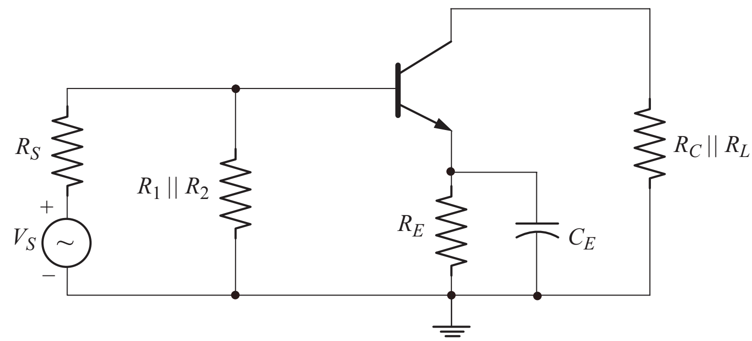


Fig. 4.17 *AC equivalent circuit to study the effect of C_E*

Let us replace the transistor by its low frequency small signal hybrid model as shown in Fig. 4.18.

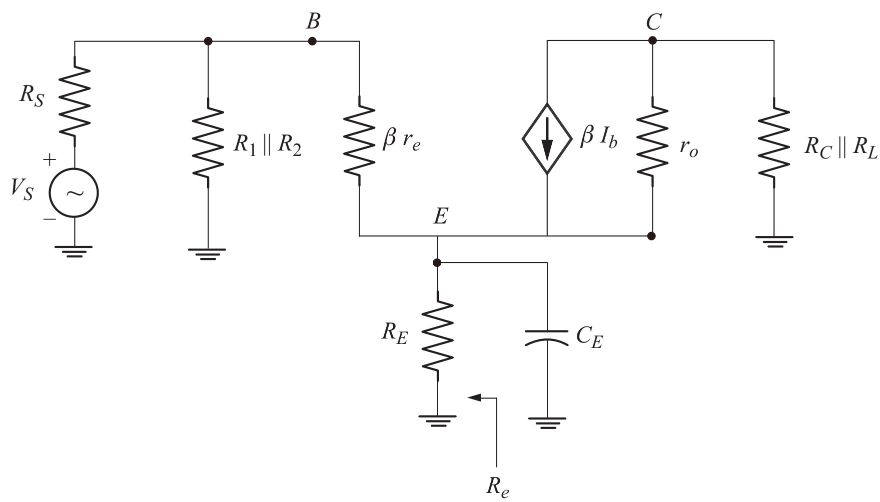


Fig. 4.18 *AC equivalent circuit using hybrid model*

R_e is the ac equivalent resistance seen by C_E. To find R_e we reduce V_S to zero as shown in Fig. 4.19. For simplicity the output circuit is omitted.

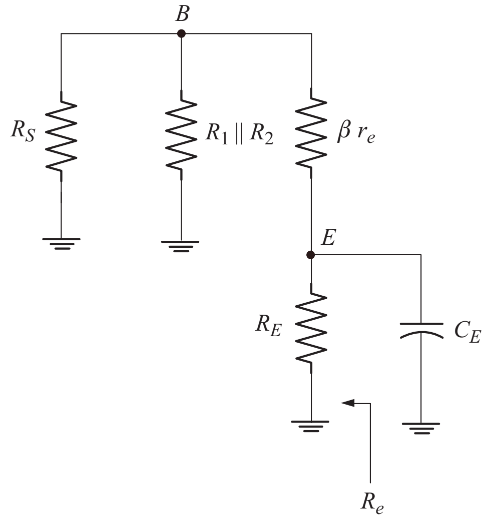


Fig. 4.19 Circuit to find R_e

$$\text{Let} \qquad R'_S = R_S \| R_1 \| R_2 \tag{4.35}$$

The ac equivalent circuit is redrawn as shown in Fig. 4.20.

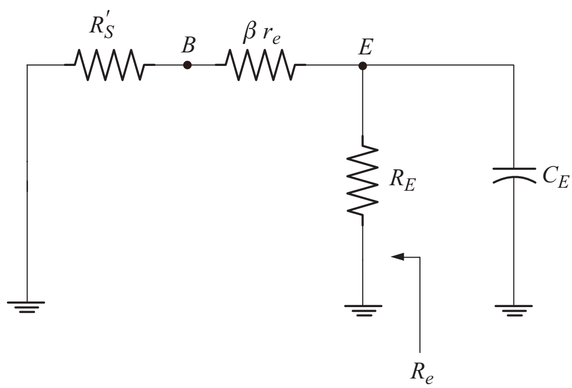


Fig. 4.20 *AC equivalent circuit redrawn*

Note that $R'_S + \beta\, r_e$ is in the base circuit. When it is transferred to the emitter circuit it gets divided by β, since the emitter current is approximately β times the base current. The resulting circuit is shown in Fig. 4.21.

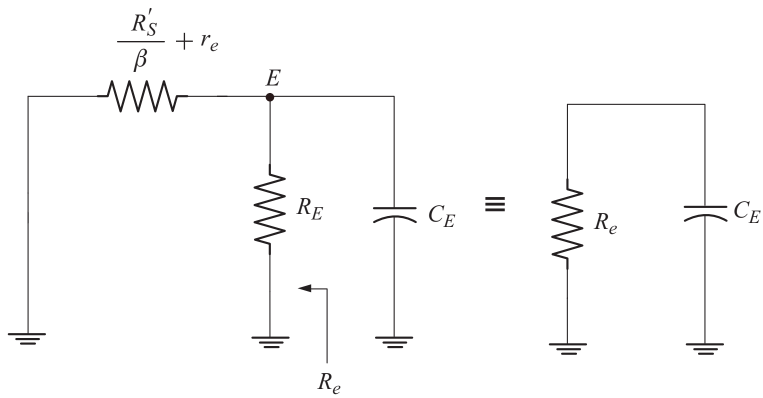


Fig. 4.21 Circuit to find R_e

Note that

$$R_e = R_E \| \left[\frac{R'_S}{\beta} + r_e \right] \tag{4.36}$$

For the circuit of Fig. 4.21, the lower cut-off frequency due to C_E is given by

$$f_{L_E} = \frac{1}{2\pi R_e C_E} \tag{4.37}$$

Effect of C_E on Voltage Gain

The midband voltage gain of the amplifier of Fig. 4.13 with out C_E is given by

$$A_{V\text{mid}} = -\frac{R_o \| R_L}{r_e + R_E} \tag{4.38}$$

where $$R_o = R_C \| r_o$$

If C_E is connected in parallel with R_E, then the voltage gain becomes a function of frequency. Now the voltage gain at any frequency is given by

$$A_V = -\frac{R_o \| R_L}{r_e + R_E \| X_{C_E}} \tag{4.39}$$

where $$X_{C_E} = \frac{1}{2\pi f C_E} \tag{4.40}$$

As the frequency increases:

- X_{C_E} decreases
- $R_E \| X_{C_E}$ decreases
- A_V increases in magnitude

As the frequency approaches the midband value

- X_{C_E} approaches zero
- $R_E \parallel X_{C_E}$ approaches zero (i.e R_E is shorted out)
- A_V approaches the maximum value or midband value

$$A_{V\,\text{mid}} = -\frac{R_o \parallel R_L}{r_e} \tag{4.41}$$

Note that, Equation (4.41) can be obtained from Equation (4.39) by substituting $R_E = 0\ \Omega$

Over all Lower Cut-off Frequency

The low frequency response of the amplifier is influenced by the capacitors C_S, C_C and C_E. The lower cut-off frequencies due to C_S, C_C and C_E respectively are f_{L_S}, f_{L_C} and f_{L_E}. If these cut-off frequencies are relatively apart (i.e., one is greater than the other by four times or more) the higher of the three is approximately the lower cut-off frequency for the amplifier stage.

For example if $f_{L_S} = 6\text{ Hz}$, $f_{L_C} = 25\text{ Hz}$ and $f_{L_E} = 320\text{ Hz}$

then the lower cut-off frequency of the amplifier is $f_{L_E} = 320$ Hz, since

$$f_{L_C} > 4 f_{L_S} \quad \text{and} \quad f_{L_E} > 4 f_{L_C}$$

Example 4.5

For the circuit shown below calculate the following:

(a) r_e

(b) Input resistance, R_i

(c) Mid band voltage gains $A_V = \dfrac{V_o}{V_i}$ and $A_{V_S} = \dfrac{V_o}{V_S}$

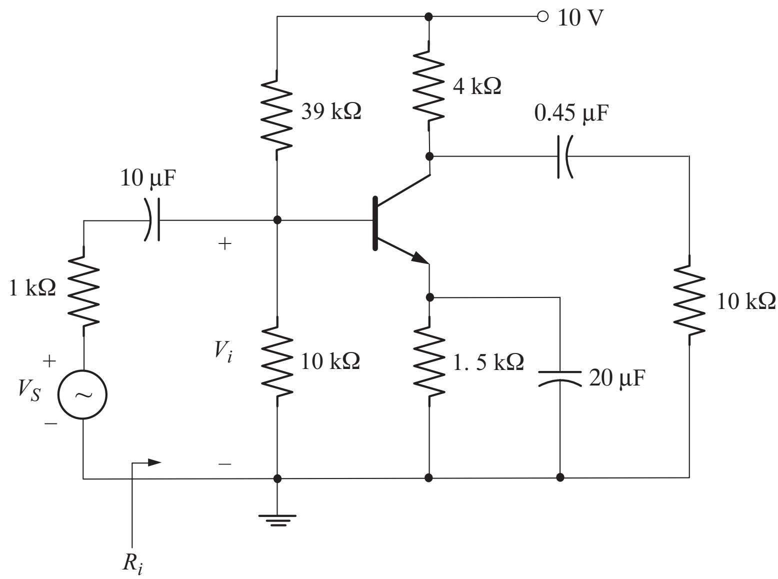
(d) Lower cut-off frequency due to C_S

(e) Lower cut-off frequency due to C_C

(f) Lower cut-off frequency due to C_E

(g) Overall lower cut-off frequency

For the transistor, $\beta = 100$ and $r_o = \infty\ \Omega$.



Solution

(a) Calculation of r_e

Check for, $$\beta R_E \geq 10 R_2$$

$$\beta R_E = (100)(1.5\ \text{k}\Omega) = 150\ \text{k}\Omega$$

$$10 R_2 = (10)(10\ \text{k}\Omega) = 100\ \text{k}\Omega$$

Since $\beta R_E > 10 R_2$, we can use approximate analysis

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{(10\ \text{V})(10\ \text{k}\Omega)}{39\ \text{k}\Omega + 10\ \text{k}\Omega} = 2.04\ \text{V}$$

$$V_E = V_B - V_{BE} = 2.04\ \text{V} - 0.7\ \text{V} = 1.34\ \text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.34\ \text{V}}{1.5\ \text{k}\Omega} = 0.893\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{0.893\ \text{mA}} = 29.1\ \Omega$$

(b) ***Input Resistance***

$$R_i = R_1 \parallel R_2 \parallel \beta r_e$$

$$= 39\ \text{k}\Omega \parallel 10\ \text{k}\Omega \parallel (100)(29.1\ \Omega) = 2.13\ \text{k}\Omega$$

(c) ***Midband Voltage Gain***

$$A_V = \frac{V_o}{V_i} = -\frac{R_o \parallel R_L}{r_e}$$

$$R_o = R_C \,||\, r_o = 4\text{ k}\Omega \,||\, \infty = 4\text{ k}\Omega$$

$$R_o \,||\, R_L = 4\text{ k}\Omega \,||\, 10\text{ k}\Omega = 2.86\text{ k}\Omega$$

$$A_V = -\frac{2.86\text{ k}\Omega}{29.1\ \Omega} = -98.28$$

$$A_{V_S} = \frac{V_o}{V_S}$$

$$= \frac{V_o}{V_i}\cdot\frac{V_i}{V_S}$$

$$= A_V\frac{V_i}{V_S} \qquad \text{(A)}$$

In the midband, C_S represents short circuit. The input equivalent circuit in the mid band is shown in Fig. A.

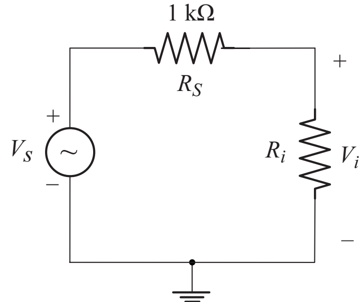


Fig. A

Using voltage division rule

$$V_i = \frac{V_S R_i}{R_S + R_i}$$

$$\frac{V_i}{V_S} = \frac{R_i}{R_S + R_i} = \frac{2.13\text{ k}\Omega}{1\text{ k}\Omega + 2.13\text{ k}\Omega} = 0.68$$

Using this value in Equation (A) we get

$$A_V = (-98.28)(0.68) = -66.83$$

(d) *Lower Cut-off Frequency due to C_S*

$$f_{L_S} = \frac{1}{2\pi\left[R_S + R_i\right]C_S}$$

$$= \frac{1}{2\pi\left[1\text{ k}\Omega + 2.13\text{ k}\Omega\right]\left[10\ \mu\text{F}\right]} = 5.08\text{ Hz}$$

(e) ***Lower Cut-off Frequency due to C_C***

$$f_{L_C} = \frac{1}{2\pi\left[R_o + R_L\right]C_C}$$

$$R_o = R_C \,||\, r_o = 4\text{ k}\Omega \,||\, \infty = 4\text{ k}\Omega$$

$$f_{L_C} = \frac{1}{2\pi\left[4\text{ k}\Omega + 10\text{ k}\Omega\right]\left[0.45\ \mu\text{F}\right]} = 25.26\text{ Hz}$$

(f) ***Lower Cut-off frequency due to C_E***

$$f_{L_E} = \frac{1}{2\pi R_e C_E}$$

$$R_e = R_E \,||\, \left[\frac{R_S'}{\beta} + r_e\right]$$

$$R_S' = R_S \,||\, R_1 \,||\, R_2 = 1\text{ k}\Omega \,||\, 39\text{ k}\Omega \,||\, 10\text{ k}\Omega = 0.888\text{ k}\Omega$$

$$\frac{R_S'}{\beta} + r_e = \frac{0.888\text{ k}\Omega}{100} + 29.1\ \Omega = 37.98\ \Omega$$

$$R_e = 1.5\text{ k}\Omega \,||\, 37.98\ \Omega = 37\ \Omega$$

$$f_{L_E} = \frac{1}{2\pi\left[37\,\Omega\right]\left[20\ \mu\text{F}\right]} = 215\text{ Hz}$$

(g) ***Over all Lower Cut-off Frequency***

We have

$$f_{L_S} = 5.08\text{ Hz} \quad f_{L_C} = 25.26\text{ Hz} \quad f_{L_E} = 215\text{ Hz}$$

Note that $\quad f_{L_C} > 4 f_{L_S}$

and $\quad f_{L_E} > 4 f_{L_C}$

Therefore over all lower cut-off frequency is f_{LE}

$$f_L = f_{L_E} = 215\text{ Hz}$$

Example 4.6

Repeat example 4.5 with $r_o = 40\text{ k}\Omega$.

Solution

$$r_o = 40\text{ k}\Omega$$

$$R_C = 4\text{ k}\Omega$$

Since $r_o \geq 10\ R_C$, the effect of r_o can be neglected. Therefore, all the results remain unaffected.

Example 4.7

Repeat example 4.5 with $r_o = 20\ \text{k}\Omega$.

Solution

In this case $r_o < 10\ R_C$. Only A_V, $A_{V\,\text{mid}}$ and f_{L_C} get affected. Other values remain unchanged.

$$\left.\begin{array}{ll} r_e = 29.1\ \Omega & R_i = 2.13\ \text{k}\Omega \\ f_{L_S} = 5.08\ \text{Hz} & f_{L_E} = 215\ \text{Hz} \end{array}\right\} \text{As calculated in example 4.5}$$

Midband Voltage Gain

$$A_V = \frac{V_o}{V_i} = -\frac{R_o \parallel R_L}{r_e}$$

$$R_o = R_C \parallel r_o = 4\text{k}\ \Omega \parallel 20\ \text{k}\Omega = 3.33\ \text{k}\Omega$$

$$R_o \parallel R_L = 3.33\ \text{k}\Omega \parallel 10\ \text{k}\Omega = 2.49\ \text{k}\Omega$$

$$A_V = -\frac{2.49\ \text{k}\Omega}{29.1\ \Omega} = -85.56$$

$$A_{V_S} = A_V\left[\frac{R_i}{R_i + R_S}\right] = (-85.56)\,(0.68) = -58.18$$

Lower Cut-off Frequency due to C_C

$$f_{L_C} = \frac{1}{2\pi\left[R_o + R_L\right]C_C}$$

$$= \frac{1}{2\pi[3.33\ \text{k}\Omega + 10\ \text{k}\Omega][0.45\ \mu\text{F}]} = 26.53\ \text{Hz}$$

Over all Lower Cut-off Frequency

$$f_{L_S} = 5.08\ \text{Hz}$$

$$f_{L_C} = 26.53\ \text{Hz}$$

$$f_{L_E} = 215\ \text{Hz}$$

Note that $f_{L_C} > 4 f_{L_S}$

and $f_{L_E} > 4 f_{L_S}$

Hence the overall lower cut-off frequency is

$$f_L = f_{L_E} = 215\ \text{Hz}$$

Example 4.8

For the circuit shown below calculate the following:

(a) r_e

(b) Input resistance R_i

(c) Mid band voltage gains $A_V = \frac{V_o}{V_i}$ and $A_{V_S} = \frac{V_o}{V_s}$

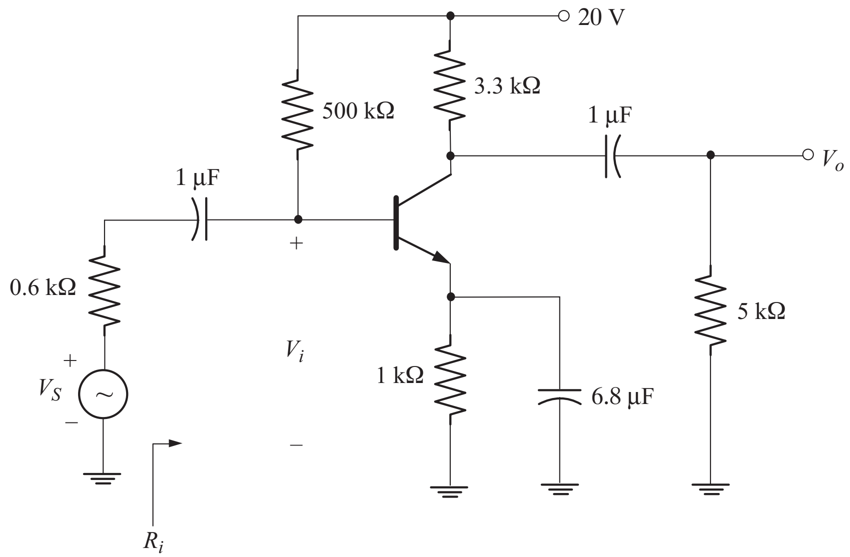
(d) Lower cut-off frequency due to C_S

(e) Lower cut-off frequency due to C_C

(f) Lower cut-off frequency due to C_E

(g) Over all lower cut-off frequency

For the transistor $\beta = 100$ and $r_o = \infty$.



Solution

(a) *Calculation of r_e*

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1+\beta)R_E}$$

$$= \frac{20\,\text{V} - 0.7\,\text{V}}{500\text{ k}\Omega + (101)(1\text{ k}\Omega)} = 32.11\ \mu\text{A}$$

$$I_E = (1+\beta)\, I_B = (101)\,(32.11\ \mu\text{A}) = 3.24\text{ mA}$$

$$r_e = \frac{26\,\text{mV}}{I_E} = \frac{26\,\text{mV}}{3.24\,\text{mA}} = 8.02\ \Omega$$

(b) *Input Resistance*

$$R_i = R_B \| \beta\, r_e$$

$$= 500\text{ k}\Omega \,\|\, (100)\,(8.02\ \Omega) = 800.72\ \Omega$$

(c) ***Midband Voltage Gain***

$$A_V = \frac{V_o}{V_i} = -\frac{R_o \parallel R_L}{r_e}$$

$$R_o = R_C \parallel r_o = 3.3\ \text{k}\Omega \parallel \infty = 3.3\ \text{k}\Omega$$

$$R_o \parallel R_L = 3.3\ \text{k}\Omega \parallel 5\ \text{k}\Omega = 1.98\ \text{k}\Omega$$

$$A_V = -\frac{1.98\ \text{k}\Omega}{8.02\ \Omega} = -246.88$$

$$A_{V_S} = A_V \frac{R_i}{R_S + R_i}$$

$$= [-246.88]\frac{800.72\ \Omega}{0.6\ \text{k}\Omega + 800.72\ \Omega} = -141.12$$

(d) ***Lower Cut-off Frequency due to C_S***

$$f_{L_S} = \frac{1}{2\pi\left[R_S + R_i\right]C_S}$$

$$= \frac{1}{2\pi\left[0.6\ \text{k}\Omega + 800.72\ \Omega\right]\left[1\ \mu\text{F}\right]} = 113.6\ \text{Hz}$$

(e) ***Lower Cut-off Frequency due to C_C***

$$f_{L_C} = \frac{1}{2\pi\left[R_o + R_L\right]C_C}$$

$$= \frac{1}{2\pi\left[3.3\ \text{k}\Omega + 5\ \text{k}\Omega\right]\left[1\ \mu\text{F}\right]} = 19.17\ \text{Hz}$$

(f) ***Lower Cut-off Frequency due to C_E***

$$f_{L_E} = \frac{1}{2\pi R_e C_E}$$

$$R_e = R_E \parallel \left[\frac{R'_S}{\beta} + r_e\right]$$

$$R'_S = R_B \parallel R_S = 500\ \text{k}\Omega \parallel 0.6\ \text{k}\Omega = 599.28\ \Omega$$

$$\frac{R'_S}{\beta} + r_e = \frac{599.28\Omega}{100} + 8.02\ \Omega = 14.01\ \Omega$$

$$R_e = 1\ \text{k}\Omega \parallel 14.01\ \Omega = 13.82\ \Omega$$

$$f_{L_E} = \frac{1}{2\pi\left[13.82\Omega\right]\left[6.8\mu F\right]} = 1693.5\ \text{Hz}$$

(g) *Over all Lower 3dB Frequency*

$$f_{L_S} = 113.6 \text{ Hz}$$
$$f_{L_C} = 19.17 \text{ Hz}$$
$$f_{L_E} = 1693.5 \text{ Hz}$$

Note that
$$f_{L_S} > 4\, f_{L_C}$$
$$f_{L_E} > 4\, f_{L_S}$$

Hence the over all lower 3 dB frequency is

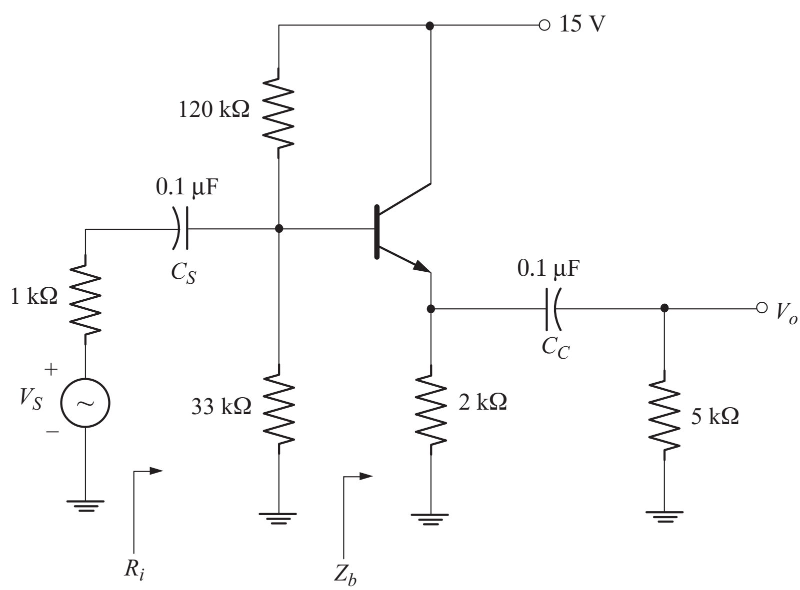
$$f_L = f_{L_E} = 1693.5 \text{ Hz}$$

Example 4.9

For the emitter follower circuit shown below, calculate the following:

(a) r_e
(b) Input resistance R_i
(c) Mid band voltage gains $A_V = \dfrac{V_o}{V_i}$ and $A_{V_S} = \dfrac{V_o}{V_S}$
(d) Lower cut-off frequency due to C_S
(e) Lower cut-off frequency due to C_C
(f) Over all lower cut-off frequency

For the transistor, $\beta = 100$ and $r_o = \infty$.



Solution

(a) ***Calculation of r_e***

$$\beta R_E = (100)(2\ \text{k}\Omega) = 200\ \text{k}\Omega$$

$$10 R_2 = (10)(33\ \text{k}\Omega) = 330\ \text{k}\Omega$$

Note that, $$\beta R_E < 10 R_2$$

Hence we have to use exact analysis

$$V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{(15\ \text{V})(33\ \text{k}\Omega)}{120\ \text{k}\Omega + 33\ \text{k}\Omega} = 3.23\ \text{V}$$

$$R_{Th} = R_1 \parallel R_2 = 120\ \text{k}\Omega \parallel 33\ \text{k}\Omega = 25.88\ \text{k}\Omega$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (1+\beta)R_E} = \frac{3.23\ \text{V} - 0.7\ \text{V}}{25.88\ \text{k}\Omega + (101)(2\ \text{k}\Omega)} = 11.1\ \mu\text{A}$$

$$I_E = (1+\beta)\, I_B = (101)(11.1\ \mu\text{A}) = 1.12\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{1.12\ \text{mA}} = 23.21\ \Omega$$

(b) ***Input Resistance***

$$R_i = R_1 \parallel R_2 \parallel Z_b$$

$$Z_b = \beta r_e + (1+\beta)[R_E \parallel R_L]$$

$$= (100)(23.21\Omega) + (101)[2\text{k}\Omega \parallel 5\text{k}\Omega] = 146.6\ \text{k}\Omega$$

$$R_i = 120\ \text{k}\Omega \parallel 33\ \text{k}\Omega \parallel 146.6\ \text{k}\Omega = 22\ \text{k}\Omega$$

(c) ***Midband Voltage Gain***

$$A_V = \frac{R_E \parallel R_L}{(R_E \parallel R_L) + r_e}$$

$$R_E \parallel R_L = 2\ \text{k}\Omega \parallel 5\ \text{k}\Omega = 1.43\ \text{k}\Omega$$

$$A_V = \frac{1.43\ \text{k}\Omega}{1.43\ \text{k}\Omega + 23.21\ \Omega} = 0.984$$

$$A_{V_S} = A_V \frac{R_i}{R_S + R_i} = (0.984)\left[\frac{22\ \text{k}\Omega}{22\ \text{k}\Omega + 1\ \text{k}\Omega}\right] = 0.941$$

(d) ***Lower Cut-off Frequency due to C_S***

$$f_{L_S} = \frac{1}{2\pi [R_S + R_i] C_S}$$

$$= \frac{1}{2\pi [1\ \text{k}\Omega + 22\ \text{k}\Omega][0.1\ \mu\text{F}]} = 69.19\ \text{Hz}$$

(e) *Lower Cut-off Frequency due to C_C*

$$f_{L_C} = \frac{1}{2\pi\left[R_o + R_L\right]C_C}$$

$$R_o = R_E \parallel r_e = 2\text{ k}\Omega \parallel 23.21\ \Omega = 22.94\ \Omega$$

$$f_{L_C} = \frac{1}{2\pi\left[22.94\ \Omega + 5\text{ k}\Omega\right]\left[0.1\ \mu\text{F}\right]} = 316.85\text{ Hz}$$

(g) *Over all Lower Cut-off Frequency*

$$f_{L_S} = 66.39\text{ Hz} \quad f_{L_C} = 316.85\text{ Hz}$$

Note that $f_{L_C} > 4 f_{L_S}$

Hence f_{L_C} is the over all lower cut-off frequency.

Example 4.10

For the common-base amplifier shown below, calculate the following:

(a) r_e

(b) Input resistance, R_i

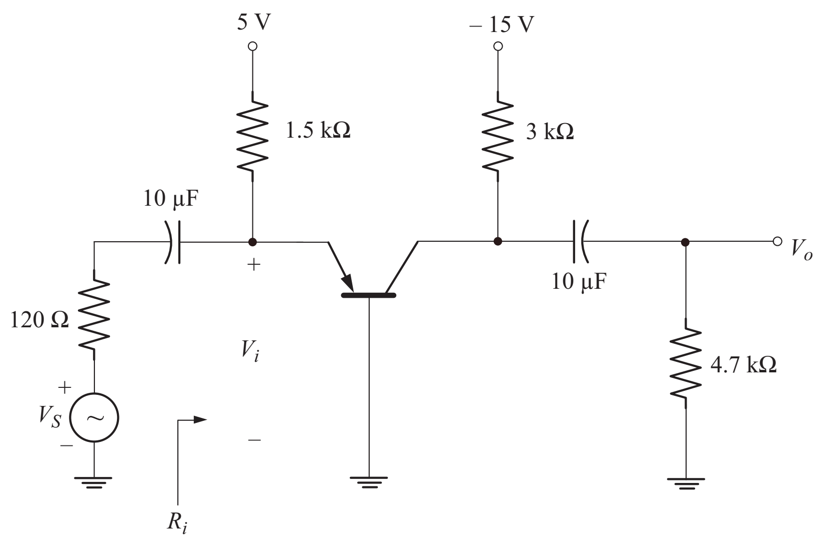
(c) Mid band voltage gains $A_V = \dfrac{V_o}{V_i}$ and $A_{V_S} = \dfrac{V_o}{V_S}$

(d) Lower cut-off frequency due to C_S

(e) Lower cut-off frequency due to C_C

(f) Over all lower cut-off frequency.

For the transistor $\beta = 75$ and $r_o = \infty$.



Solution

(a) ***Calculation of r_e***

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{5\text{ V} - 0.7\text{ V}}{1.5\text{ k}\Omega} = 2.86\text{ mA}$$

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{2.86\text{ mA}} = 9.09\ \Omega$$

(b) ***Input Resistance***

$$R_i = R_E \,||\, r_e = 1.5\text{ k}\Omega \,||\, 9.09\ \Omega = 9.03\ \Omega$$

(c) ***Midband Voltage Gain***

$$A_V = \frac{\alpha\left[R_C \,||\, R_L\right]}{r_e}$$

$$\alpha = \frac{\beta}{1+\beta} = \frac{75}{76} = 0.986$$

$$R_C \,||\, R_L = 3\text{ k}\Omega \,||\, 4.7\text{ k}\Omega = 1.83\text{ k}\Omega$$

$$A_V = \frac{(0.986)(1.83\text{ k}\Omega)}{9.09\ \Omega} = 198.5$$

$$A_{V_S} = A_V\left[\frac{R_i}{R_S + R_i}\right] = [198.5]\left[\frac{9.03\ \Omega}{120\ \Omega + 9.03\ \Omega}\right] = 13.89$$

(d) ***Lower Cut-off Frequency due to C_S***

$$f_{L_S} = \frac{1}{2\pi\left[R_S + R_i\right]C_S}$$

$$= \frac{1}{2\pi\left[120\ \Omega + 9.03\ \Omega\right]\left[10\ \mu\text{F}\right]} = 123.34\text{ Hz}$$

(e) ***Lower cut-off Frequency due to C_C***

$$f_{L_C} = \frac{1}{2\pi\left[R_o + R_L\right]C_C}$$

$$R_o = R_C = 3\text{ k}\Omega$$

$$f_{L_C} = \frac{1}{2\pi\left[3\text{ k}\Omega + 4.7\text{ k}\Omega\right]\left[10\ \mu\text{F}\right]} = 2.06\text{ Hz}$$

(f) ***Over all Lower Cut-off Frequency***

$$f_{L_S} = 123.34\text{ Hz}$$

$$f_{L_C} = 2.06\text{ Hz}$$

Note that $$f_{L_S} > 4f_{L_C}$$

$\therefore$ Over all lower cut-off frequency is f_{L_S}

$$f_L = f_{L_S} = 123.34 \text{ Hz}$$

Example 4.11

It is desired that the voltage gain of an RC-coupled amplifier at 60 Hz should not decrease by more than 10 % from its mid band value. Show that the coupling capacitor C_S must be at least equal to $\frac{5.5}{R}$ where $R = R_S + R_i$. R is in kilo-ohms and C_S in micro farads. Neglect the effects of C_E and C_C.

Solution

(a) Voltage gain in low frequency region is given by

$$|A_V| = \frac{1}{\sqrt{1+\left(\frac{f_L}{f}\right)^2}} \tag{A}$$

at $f = 60$ Hz, $$|A_V| > 0.9$$

Using this data in Equation (A), we have

$$\frac{1}{\sqrt{1+\left(\frac{f_L}{60}\right)^2}} > 0.9$$

$$\Rightarrow \quad 1+\left(\frac{f_L}{60}\right)^2 < 1.235$$

or $$f_L < 29 \text{ Hz}$$

(b) Since the effects of C_C and C_E are to be neglected,

$$f_L = f_{L_S}$$

But $$f_{L_S} = \frac{1}{2\pi\,[R_S + R_i]\,C_S} < 29 \text{ Hz}$$

$$2\pi\,[R_S + R_i]\,C_S > \frac{1}{29}$$

$$C_S > \frac{1}{(2\pi)(29)(R_S + R_i)}$$

Since C_S is in micro Farad and $R = R_S + R_i$ in kilo-ohms we have

$$C_S\,[10^{-6}] > \frac{1}{(2\pi)(29)(R)(10^3)}$$

$$C_S > \frac{5.5}{R}$$

Example 4.12

It is desired that the lower cut-off frequency due to C_S be not more than 10 Hz for the RC-coupled amplifier of Fig. 4.13. Calculate the minimum value of input coupling capacitor C_S. Take R_1 = 39 kΩ, R_2 = 10 kΩ, R_S =1 kΩ, β = 100, V_{CC} = 10 V.

Solution

Given $$f_{L_S} < 10\text{Hz}$$

$$f_{L_S} = \frac{1}{2\pi\,[R_S + R_i]\,C_S} < 10\text{ Hz}$$

$$2\,\pi\,[R_S + R_i]\,C_S > 0.1$$

$$C_S > \frac{0.1}{2\pi\,[R_S + R_i]}$$

$$R_i = 2.13\text{ k}\Omega \qquad \text{[see example 4.5]}$$

$$C_S > \frac{0.1}{2\,\pi[1\,\text{k}\Omega + 2.13\,\text{k}\Omega]}$$

$$C_S > 5.08\ \mu\text{F}$$

◆ 4.4 MILLER EFFECT CAPACITANCE

Figure 4.22 shows an inverting amplifier with a capacitance C_f between the input and output nodes.

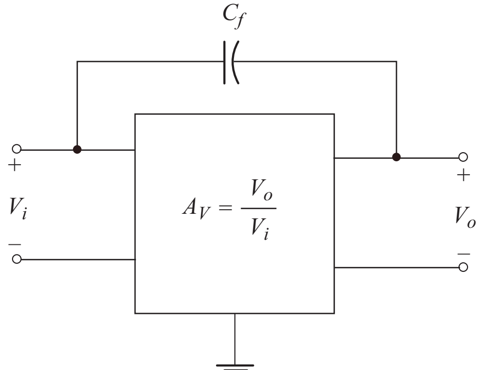


Fig. 4.22 Inverting amplifier with capacitance between input and output nodes

It is important to note that, A_V is negative for inverting amplifier since V_o and V_i are 180° out of phase. Using Millers theorem we can find the loading effect of C_f on the input and output circuits of the amplifier.

To find the Miller Input Capacitance $[C_{M_i}]$

We use the circuit of Fig. 4.23 to find the Miller input capacitance.

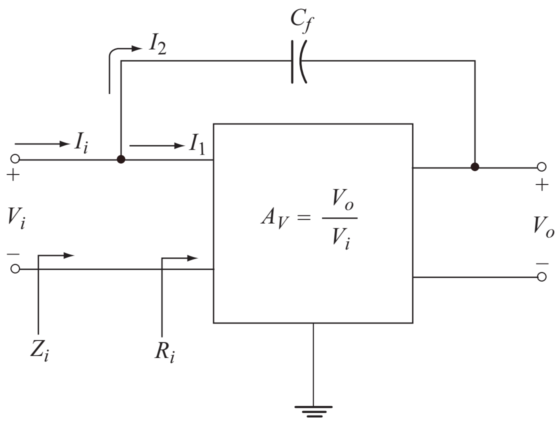


Fig. 4.23 Circuit to find Miller input capacitance

Let $$R_i = \frac{V_i}{I_1} \Rightarrow I_1 = \frac{V_i}{R_i}$$

and $$Z_i = \frac{V_i}{I_i} \Rightarrow I_i = \frac{V_i}{Z_i}$$

Applying KCL at the input node we have

$$I_i = I_1 + I_2 \tag{4.42}$$

$$I_2 = \frac{V_i - V_o}{X_{C_f}}$$

$$= \frac{V_i - A_V V_i}{X_{C_f}} \qquad [\because V_o = A_V V_i]$$

$$I_2 = \frac{[1 - A_V] V_i}{X_{C_f}}$$

Substituting for I_i, I_1 and I_2 in Equation (4.42) we get

$$\frac{V_i}{Z_i} = \frac{V_i}{R_i} + \frac{[1 - A_V] V_i}{X_{C_f}}$$

Eliminating V_i throughout we have

$$\frac{1}{Z_i} = \frac{1}{R_i} + \frac{1}{\left[\dfrac{X_{C_f}}{1-A_V}\right]}$$

$$\frac{1}{Z_i} = \frac{1}{R_i} + \frac{1}{X_{C_{Mi}}} \tag{4.43}$$

where $$X_{C_{Mi}} = \frac{X_{C_f}}{1-A_V} \tag{4.44}$$

$$X_{C_f} = \frac{1}{2\pi f C_f}$$

Using this relation in Equation (4.44) we have

$$X_{C_{Mi}} = \frac{1}{2\pi f[1-A_V]C_f}$$

$$X_{C_{Mi}} = \frac{1}{2\pi f C_{Mi}} \tag{4.45}$$

where $$C_{Mi} = [1-A_V]\, C_f \tag{4.46}$$

C_{Mi} is called the Miller input capacitance.

From Equation (4.43), Z_i can be interpreted as the impedance resulting from the parallel combination of R_i and C_{Mi}. This is shown in Fig. 4.24.

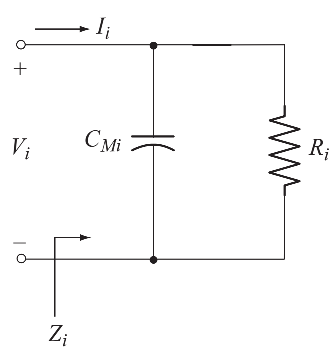


Fig 4.24 Miller input capacitance [C_{Mi}]

To find the Miller Output Capacitance [C_{Mo}]

We use the circuit of Fig. 4.25 to find the miller output capacitance.

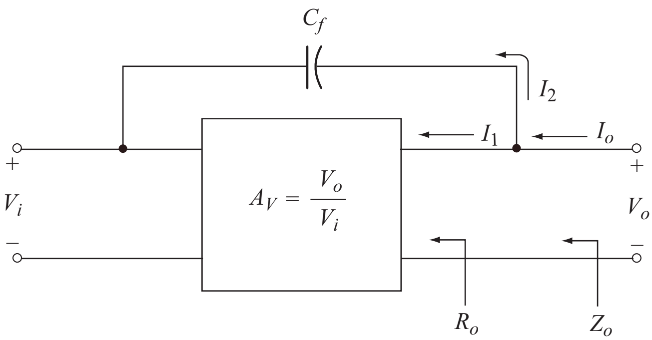


Fig 4.25 Circuit to find miller output capacitance

Let $$R_o = \frac{V_o}{I_1} \Rightarrow I_1 = \frac{V_o}{R_o}$$

and $$Z_o = \frac{V_o}{I_o} \Rightarrow I_o = \frac{V_o}{Z_o}$$

Appling KCL at the output node we have

$$I_o = I_1 + I_2 \tag{4.47}$$

$$I_2 = \frac{V_o - V_i}{X_{C_f}}$$

Using $V_i = \frac{V_o}{A_V}$ we have

$$I_2 = \frac{V_o - \frac{V_o}{A_V}}{X_{C_f}} = \frac{V_o\left[1 - \frac{1}{A_V}\right]}{X_{C_f}}$$

Substituting for I_1 and I_2 into Equation (4.47) we have

$$I_o = \frac{V_o}{R_o} + \frac{V_o\left[1 - \frac{1}{A_V}\right]}{X_{C_f}}$$

Usually R_o is large and hence the term $\frac{V_o}{R_o}$ can be neglected.

Now $$I_o \approx \frac{V_o\left[1 - \frac{1}{A_V}\right]}{X_{C_f}}$$

$$Z_o = \frac{V_o}{I_o} = \frac{X_{C_f}}{\left[1-\frac{1}{A_V}\right]}$$

$$Z_o = \frac{1}{2\pi f\left[1-\frac{1}{A_V}\right]C}$$

$$Z_o = \frac{1}{2\pi f\, C_{Mo}} \tag{4.48}$$

where
$$C_{Mo} = \left[1-\frac{1}{A_V}\right]C_f \tag{4.49}$$

C_{Mo} is called the miller output capacitance.

Summary of Millers Theorem

A capacitance C_f connected between the input and output nodes of an inverting amplifier can be replaced by

- A miller input capacitance, $C_{Mi} = [1 - A_V]\, C_f$ connected between input node and ground and
- A miller output capacitance, $C_{Mo} = \left[1-\frac{1}{A_V}\right]C_f$ connected between output node and ground.

For non inverting amplifier A_V is positive. In order to obtain positive values for C_{Mi} and C_{Mo}, Equations (4.46) and (4.49) should be modified as follows

$$C_{Mi} = [1 + A_V]\, C_f \tag{4.50}$$

$$C_{Mo} = \left[1+\frac{1}{A_V}\right]C_f \tag{4.51}$$

Application of millers theorem to the amplifier of Fig. 4.25 results in the network shown in Fig. 4.26.

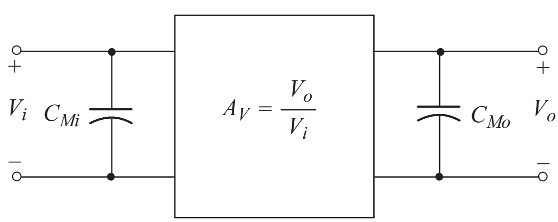


Fig. 4.26 Amplifier with C_f replaced by Miller capacitances

Applications of Millers Theorem

Millers theorem is applied:

- In feed back amplifier circuits to replace the feed back resistors / impedances connected between input and output nodes with their miller input and output impedances. An example is a voltage shunt feed back amplifier discussed in Chapter 6.
- In high frequency analysis of amplifiers to replace the internal capacitances of active devices with their miller input and output capacitances. This application is discussed in the next section.

◆ 4.5 HIGH-FREQUENCY RESPONSE OF BJT AMPLIFIER

In the high frequency response of BJT amplifier, the upper 3-dB cut-off point is defined by the following two factors.

- The network capacitance which includes the parasitic capacitances of the transistor and the wiring capacitances.
- The frequency dependence of the short circuit common emitter current gain h_{fe} or β.

4.5.1 Network Parameters

Figure 4.27 shows the RC-coupled amplifier with parasitic and wiring capacitances. C_{be}, C_{bc} and C_{ce} are the parasitic capacitances of the transistor. C_{Wi} and C_{Wo} are the input and output wiring capacitances which are introduced during the construction of the amplifier circuit.

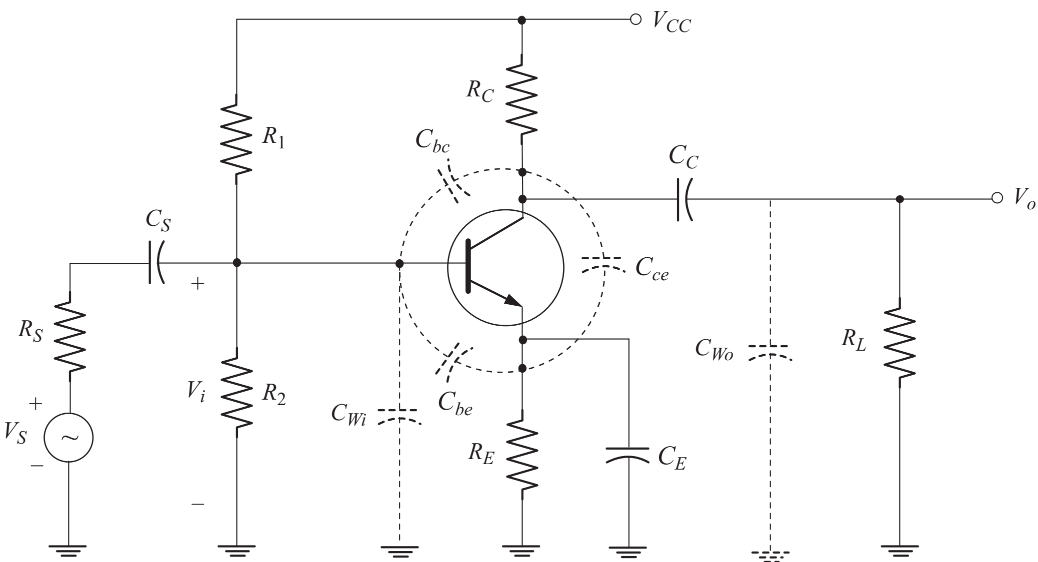


Fig. 4.27 RC-coupled amplifier with parasitic and wiring capacitances

Figure 4.28 shows the high frequency ac equivalent circuit of the RC-coupled amplifier.

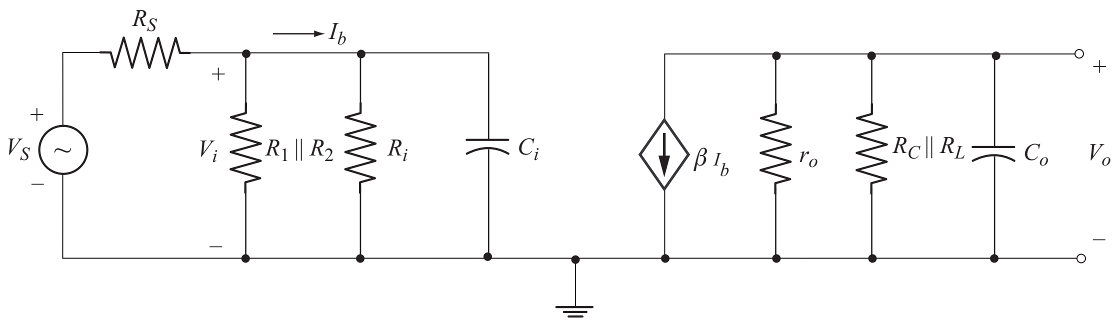


Fig. 4.28 High frequency ac equivalent circuit of the amplifier of Fig 4.27

- Using Millers theorem, the transition capacitance, C_{bc} can be replaced by two capacitances: C_{M_i} at the input and C_{M_o} at the output.
- The total input capacitance C_i, is the sum of the miller input capacitance C_{Mi}, base-emitter input capacitance C_{be} and the input wiring capacitance C_{Wi}

$$C_i = C_{Wi} + C_{be} + C_{Mi} \tag{4.52}$$

where
$$C_{Mi} = [1 - A_V]\,C_{bc} \tag{4.53}$$

- The total output capacitance C_o, is the sum of the miller output capacitance C_{M_o}, the collector-emitter parasitic capacitance C_{Ce} and the output wiring capacitance C_{W_i}

$$C_o = C_{W_o} + C_{ce} + C_{Mo} \tag{4.54}$$

where
$$C_{Mo} = \left[1 - \frac{1}{A_V}\right] C_{bc} \tag{4.55}$$

Upper Cut-off Frequency due to Input Capacitance C_i

To find the upper cut-off frequency due to the input capacitance C_i, we consider the input circuit of Fig. 4.28 which is shown in Fig. 4.29(a).

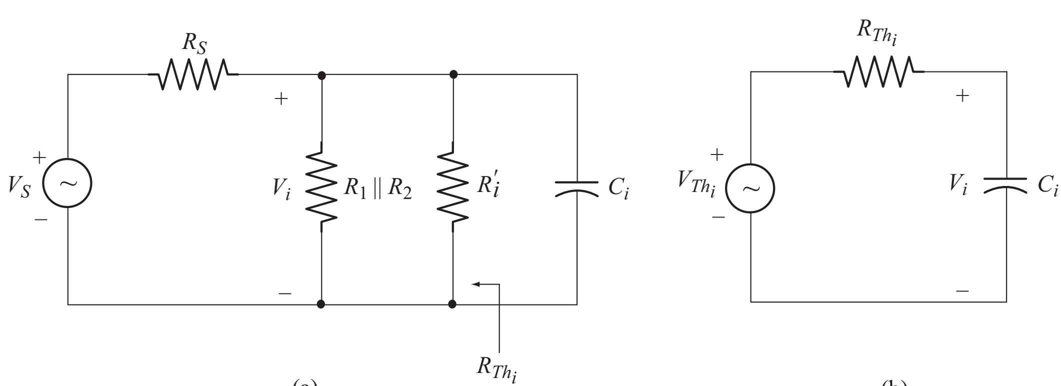


Fig. 4.29 (a) Input ac equivalent circuit (b) Thevenin equivalent of input circuit

The network consisting of V_S, R_S, $R_1 \parallel R_2$ and R_i' can be replaced by its Thevenin equivalent as shown in Fig. 4.29(b).

The Thevenin voltage V_{Th_i} and Thevenin resistance R_{Th_i} can be calculated using the networks of Fig. 4.30(a) and 4.30(b) respectively.

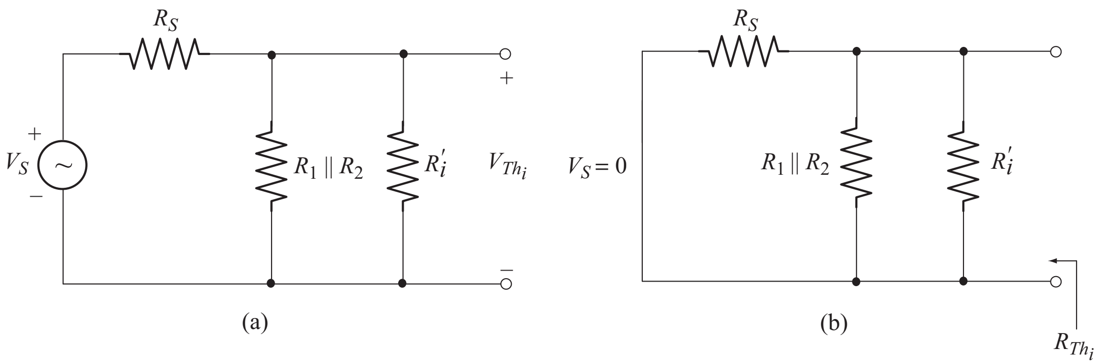


Fig. 4.30 (a) Circuit to find V_{Th_i} (b) Circuit to find R_{Th_i}

Using voltage division rule in the circuit of Fig. 4.30(a), we have

$$V_{Th_i} = V_S \left[\frac{R_1 \parallel R_2 \parallel R_i'}{R_S + R_1 \parallel R_2 \parallel R_i'} \right] \tag{4.56}$$

and from the circuit of Fig. 4.30(b)

$$R_{Th_i} = R_S \parallel R_1 \parallel R_2 \parallel R_i' \tag{4.57}$$

where $R_i' = \beta\, r_e$.

Now let us find the amplifier input voltage V_i, as a function of frequency from the circuit of Fig. 4.29(b).

Using voltage division rule we have

$$|V_i| = |V_{Th_i}| \left[\frac{X_{C_i}}{\sqrt{\left(R_{Th_i}\right)^2 + \left[X_{C_i}\right]^2}} \right]$$

$$|V_i| = \frac{|V_{Th_i}|}{\sqrt{1 + \left[\dfrac{R_{Th_i}}{X_{C_i}} \right]^2}} \tag{4.58}$$

where $$X_{C_i} = \frac{1}{2\pi f C_i} \tag{4.59}$$

- In the midband, the effect of C_i is negligible. As a result X_{C_i} can be treated as open circuit (i.e., $X_{C_i} = \infty$)

$$\therefore \qquad |V_i|_{\text{mid}} \approx |V_{Th_i}|$$

- At high frequencies, C_i comes into play. With increase in frequency, X_{C_i} decreases, $\dfrac{R_{Th_i}}{X_{C_i}}$ increases, $|V_i|$ decreases and hence the voltage gain decreases.

3dB cut-off occurs at a frequency at which

$$|V_i| = \frac{|V_i|_{\text{mid}}}{\sqrt{2}} = \frac{|V_{Th_i}|}{\sqrt{2}}$$

From Equation (4.58) we find that, this condition occurs when

$$R_{Th_i} = X_{C_i}$$

$$R_{Th_i} = \frac{1}{2\pi f C_i}$$

or

$$f = \frac{1}{2\pi R_{Th_i} C_i} \tag{4.60}$$

Equation (4.60) gives the upper 3 dB cutoff frequency due to the input capacitance C_i. Let us denote it by f_{H_i}

$$f_{H_i} = \frac{1}{2\pi R_{Th_i} C_i} \tag{4.61}$$

Consider

$$\frac{R_{Th_i}}{X_{C_i}} = \frac{R_{Th_i}}{\left[\dfrac{1}{2\pi f C_i}\right]} = f[2\pi R_{Th_i} C_i]$$

$$= \left[\frac{f}{f_{H_i}}\right]$$

Using this relation in Equation (4.58) we have

$$|V_i| = \frac{|V_{Th_i}|}{\sqrt{1+\left[\dfrac{f}{f_{H_i}}\right]^2}} \tag{4.62}$$

Based on the above relation we can write

$$|A_V| = \frac{1}{\sqrt{1+\left[\dfrac{f}{f_{H_i}}\right]^2}} \tag{4.63}$$

Figure 4.31 shows the Bode plot of $|A_V|_{dB}$ obtained using Equation (4.63).

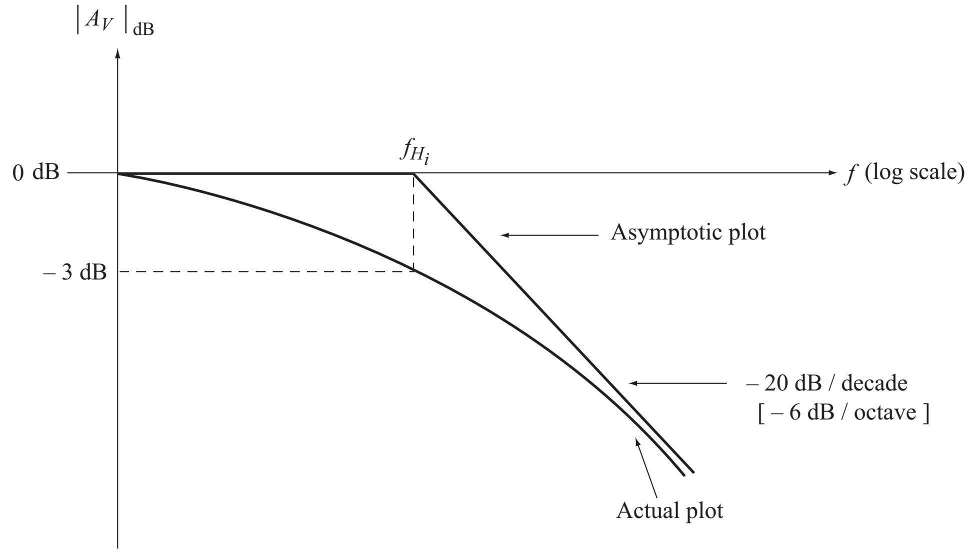


Fig. 4.31 Bode plot of equation (4.63)

Note that due to C_i, the voltage gain decreases at the rate of 20 dB / decade or 6 dB / octave in the high frequency region.

Upper Cut-off Frequency due to Output Capacitance C_o

To find the upper cut-off frequency due to output capacitance C_o, we consider the output circuit of Fig. 4.28 which is shown in Fig. 4.32.

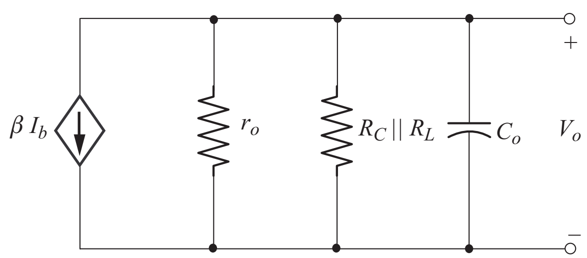


Fig. 4.32 Output ac equivalent circuit

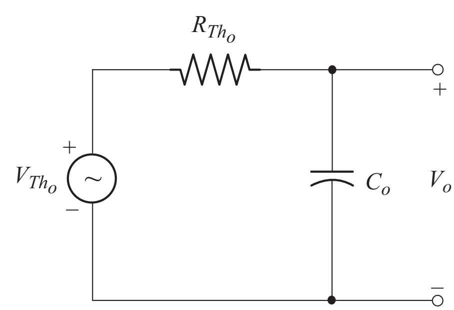


Fig. 4.33 Simplified output ac equivalent circuit

The current source βI_b along with the resistors r_o and $R_C \,||\, R_L$ can be converted into its equivalent voltage source as shown in Fig. 4.33.

$$R_{Th_o} = r_o \,||\, R_C \,||\, R_L \tag{4.64}$$

$$V_{Th_o} = [-\beta I_b]\,[r_o \,||\, R_C \,||\, R_L] \tag{4.65}$$

Using the procedure given in the previous section we can deduce the following relations for the circuit of Fig. 4.33.

The output voltage as a function of frequency is

$$|V_o| = \frac{|V_{Th_o}|}{\sqrt{1+\left[\frac{R_{Th_o}}{X_{C_o}}\right]^2}} \tag{4.66}$$

where
$$X_{C_o} = \frac{1}{2\pi f C_o} \tag{4.67}$$

The output voltage in the mid band is

$$|V_o|_{\text{mid}} \approx |V_{Th_o}| \tag{4.68}$$

The upper 3dB cutoff frequency due to C_o is

$$f_{H_o} = \frac{1}{2\,\pi\, R_{Th_o}\, C_o} \tag{4.69}$$

and the magnitude of voltage gain is

$$|A_V| = \frac{1}{\sqrt{1+\left[\frac{f}{f_{H_o}}\right]^2}} \tag{4.70}$$

Figure 4.34 shows the bode plot of Equation (4.70).

Note that due to C_o, the voltage gain decreases at the rate of 20 dB/decade or 6 dB/octave.

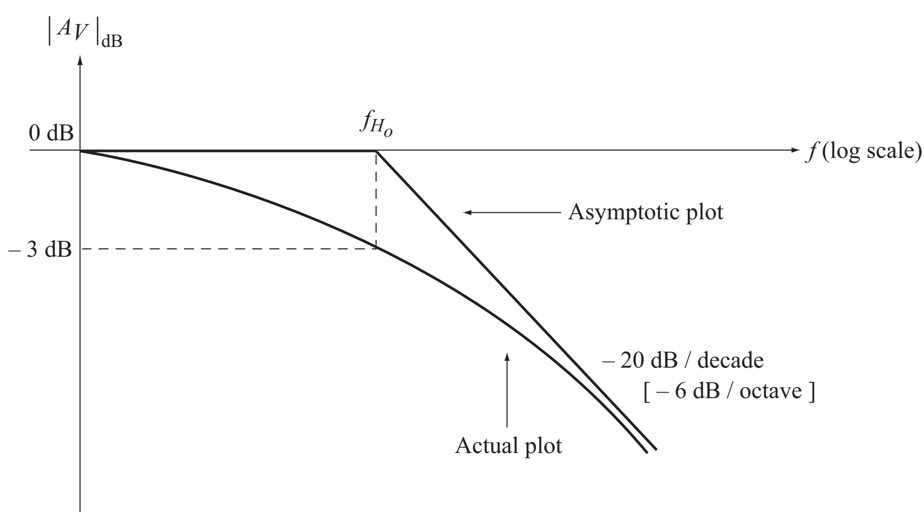


Fig. 4.34 Bode plot of equation (4.70)

Combined Effect of C_i and C_o on High Frequency Response

- The input capacitance C_i defines the upper cut-off frequency f_{H_i}
- The output capacitance C_o, defines another upper cut-off frequency f_{H_o}
- The lowest of these two frequencies will be taken as the overall upper-cutoff frequency.
- If the variation of h_{fe} with frequency is considered then the actual cut-off frequency may be lower than f_{H_i} or f_{H_o}.

4.5.2 Variation of h_{fe} (or β) with Frequency

Figure 4.35 shows the hybrid-π high frequency small signal model of BJT. Detailed explanation of this model is given in section 3.17 of Chapter 3.

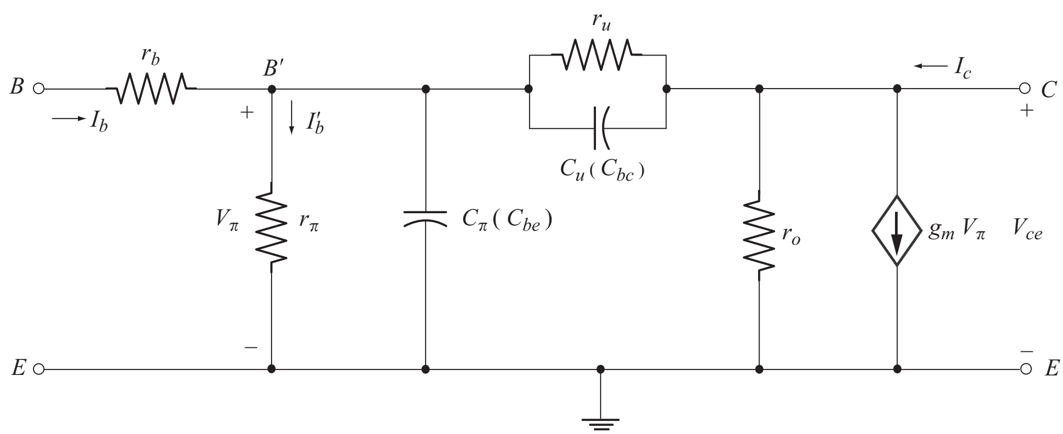


Fig. 4.35 Hybrid-π high frequency small signal model of BJT

The base-emitter input capacitance C_π (C_{be}) and the base-collector depletion capacitance C_u (C_{bc}) makes the short circuit current gain h_{fe} to vary with frequency in the high frequency region.

4.5.3 Expression for h_{fe} as a Function of Frequency

To simplify the analysis we make the following reasonable approximations:

- r_b is typically a few tens of ohms. Hence it can be treated as short circuit.
- r_u is typically a few tens of mega ohms. Hence it can be treated as open circuit.

To find the short circuit current gain, we short the output terminals. As a result r_o gets short circuited. The resulting circuit is shown in Fig. 4.36.

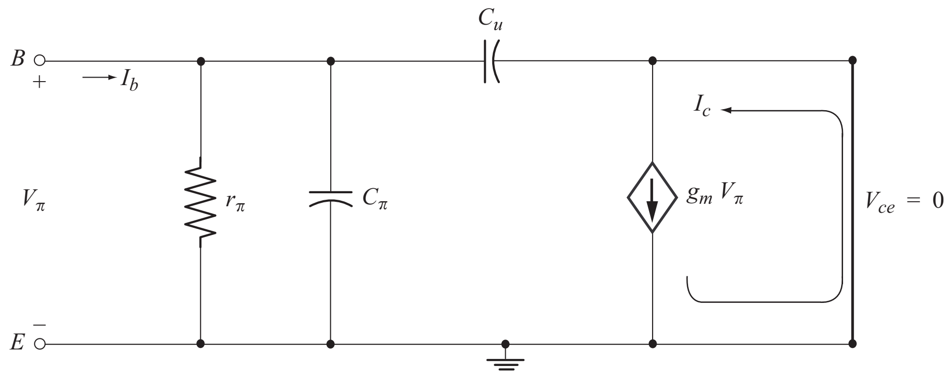


Fig. 4.36 Circuit to find short circuit current gain

Short circuit current gain is

$$h_{fe} = \left. \frac{I_c}{I_b} \right|_{V_{ce} = 0} \tag{4.71}$$

The current $g_m V_\pi$ flows into the short circuit.

$$\therefore \qquad I_c = g_m V_\pi \tag{4.72}$$

To find I_b, let us consider the input circuit of Fig. 4.36 which is shown in Fig. 4.37.

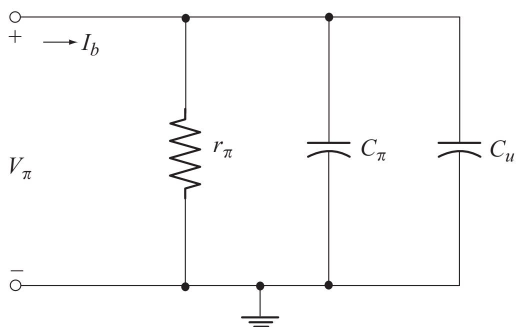


Fig. 4.37 Circuit to find I_b

Let
$$Z = r_\pi \,||\, \frac{1}{j\,\omega\,[C_\pi + C_u]} \tag{4.73}$$

$$Z = \frac{r_\pi}{1 + j\,\omega\;r_\pi\,[C_\pi + C_u]}$$

Now
$$V_\pi = I_b Z$$

$$V_\pi = \frac{I_b\, r_\pi}{1 + j\,\omega\;r_\pi\,[C_\pi + C_u]} \tag{4.74}$$

Substituting Equation (4.74) in Equation (4.72) we have

$$I_c = \frac{g_m\ r_\pi\ I_b}{1 + j\ \omega\ r_\pi \left[C_\pi + C_u\right]}$$

Now
$$h_{fe} = \frac{I_c}{I_b} = \frac{g_m\, r_\pi}{1 + j\, 2\,\pi\, f\, r_\pi \left[C_\pi + C_u\right]} \tag{4.75}$$

Let
$$f_\beta = \frac{1}{2\,\pi\ r_\pi \left[C_\pi + C_u\right]} \tag{4.76}$$

Using Equation (4.76) in Equation (4.75) we have

$$h_{fe} = \frac{g_m\, r_\pi}{1 + j\left[\dfrac{f}{f_\beta}\right]} \tag{4.77}$$

$$|\, h_{fe}| = \frac{g_m\, r_\pi}{\sqrt{1 + \left[\dfrac{f}{f_\beta}\right]^2}} \tag{4.78}$$

In the mid band, $f \ll f_\beta$. As a result $\left[\dfrac{f}{f_\beta}\right]^2 \ll 1$

From Equation (4.78) we have

$$|\, h_{fe}|_{\text{mid}} = g_m\, r_\pi = h_{fe\,\text{mid}} \tag{4.79}$$

$h_{fe\,\text{mid}}$ is also denoted by β_{mid}

Using Equation (4.79) in Equation (4.78) we have

$$|\, h_{fe}| = \frac{h_{fe\,\text{mid}}}{\sqrt{1 + \left[\dfrac{f}{f_\beta}\right]^2}} \tag{4.80}$$

Equation (4.80) gives the variation of $|\, h_{fe}\,|$ with frequency.

- As the frequency increases, $\left[\dfrac{f}{f_\beta}\right]^2$ increases and hence $|\, h_{fe}\,|$ decreases.
- When $f = f_\beta$

$$|\, h_{fe}| = \frac{h_{fe\,\text{mid}}}{\sqrt{2}} = \frac{\beta_{\text{mid}}}{\sqrt{2}}$$

Note that f_β defines the upper 3-dB cut-off point for the short circuit current gain h_{fe} . f_β is also denoted by $f_{h_{fe}}$. Now Equation (4.76) can be written as

$$f_\beta = f_{h_{fe}} = \frac{1}{2\pi\, r_\pi \left[C_\pi + C_u\right]} \tag{4.81}$$

Using $r_\pi = \beta\, r_e = \beta_{\text{mid}}\, r_e$ we have

$$f_\beta = f_{h_{fe}} = \frac{1}{2\pi\, \beta_{\text{mid}}\, r_e \left[C_\pi + C_u\right]} \tag{4.82}$$

f_β is called the β cut-off frequency. f_β is also the bandwidth for the short circuit current gain h_{fe}.

Figure 4.38 shows the variation of $|h_{fe}|$ with frequency. Note that $|h_{fe}|$ decreases from its mid band value $h_{fe\,\text{mid}}$ with a slope of 20 dB/decade or 6 dB/octave.

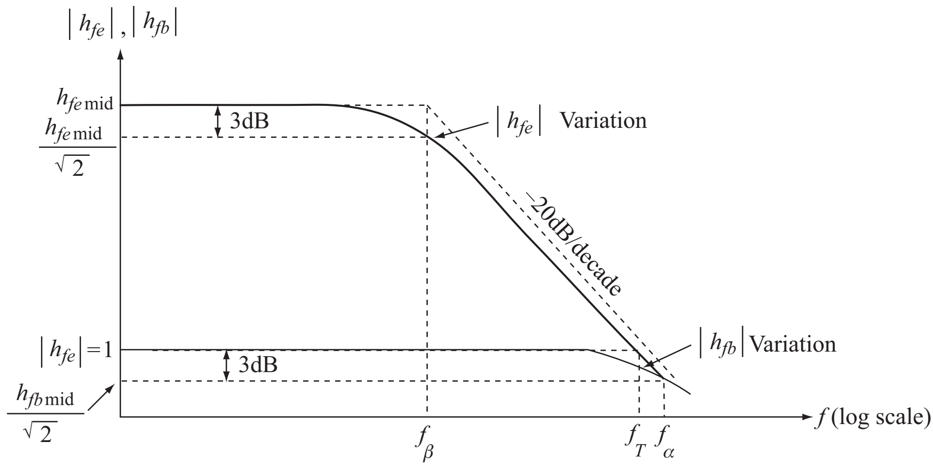


Fig. 4.38 Plot of $|h_{fe}|$ and $|h_{fb}|$ versus frequency in the high-frequency region

4.5.4 Expression for the gain-band width product [f_T]

The gain-band width product, f_T is the frequency at which $|h_{fe}| = 1$ or $|h_{fe}|_{\text{dB}} = 0$ dB.

Substituiting this condition in Equation (4.80) we have

$$\left.\frac{h_{fe\,\text{mid}}}{\sqrt{1+\left[\frac{f}{f_\beta}\right]}}\right|_{f=f_T} = 1$$

$$\frac{h_{fe\,\text{mid}}}{\sqrt{1+\left[\frac{f_T}{f_\beta}\right]^2}} = 1 \tag{4.83}$$

Since $f_T \gg f_\beta$, $$\left[\frac{f_T}{f_\beta}\right]^2 \gg 1$$

$$\therefore \qquad \sqrt{1+\left[\frac{f_T}{f_\beta}\right]^2} \approx \frac{f_T}{f_\beta}$$

Using this relation in Equation (4.83) we have

$$\frac{h_{fe\,\text{mid}}}{\left[\frac{f_T}{f_\beta}\right]} = 1$$

or $$f_T = h_{fe\,\text{mid}}\, f_\beta = \beta_{\text{mid}}\; f_\beta \tag{4.84}$$

Note that $h_{fe\,\text{mid}}$ is the midband short circuit current gain and f_β is the bandwidth.

For this reason f_T is called the gain-bandwidth product.

Substituiting Equation (4.82) into Equation (4.84) we have

$$f_T = \beta_{\text{mid}} \cdot \frac{1}{2\,\pi\,\beta_{\text{mid}}\, r_e\,[C_\pi + C_u]}$$

$$f_T = \frac{1}{2\,\pi\, r_e\,[C_\pi + C_u]} \tag{4.85}$$

4.5.5 Expression for α Cut-off Frequency [f_α]

Figure 4.38 also shows the plot of $|\,h_{fb}\,|$ versus frequency. It is important to note that $h_{fb\,\text{mid}} = 1$, since the short circuit current gain in CB configuration is unity.

α cut-off frequency, f_α is the frequency at which $|\,h_{fb}\,|$ drops to $\frac{h_{fb\,\text{mid}}}{\sqrt{2}}$. It is also frequency at which $|\,h_{fe}\,|$ drops to $\frac{1}{\sqrt{2}}$. Using this condition in Equation (4.78), it can be shown that

$$f_\beta = f_\alpha\,[\,1-\alpha\,] \tag{4.86}$$

where $$\alpha = \alpha_{\text{mid}} = h_{fb\,\text{mid}} \tag{4.87}$$

Example 4.13

Consider the circuit of Example 4.5 with the same parameter values. Calculate the following

(a) f_{H_i} and f_{H_o}

(b) f_β and f_T

Take $C_\pi\,(C_{be}) = 35$ pF $\quad C_u\,(C_{bc}) = 5$ pF $\quad C_{ce} = 1$ pF $\quad C_{Wi} = 6$ pF $\quad C_{Wo} = 10$ pF.

Solution

$$\left.\begin{aligned} r_e &= 29.1\ \Omega \\ A_V &= -99.28 \end{aligned}\right\}\ \text{From example 4.5}$$

(a)

$$f_{H_i} = \frac{1}{2\,\pi\,R_{Th_i}\,C_i}$$

$$R_{Th_i} = R_S \,\|\, R_1 \,\|\, R_2 \,\|\, R_i'$$

$$R_i' = \beta\, r_e = (100)\,(29.1\ \Omega) = 2.91\ \text{k}\Omega$$

$$R_{Th_i} = 1\ \text{k}\Omega\ \|\ 39\ \text{k}\Omega\ \|\ 10\ \text{k}\Omega\ \|\ 2.91\ \text{k}\Omega = 0.68\ \text{k}\Omega$$

$$C_i = C_{Wi} + C_{be} + C_{Mi}$$

$$= 6\ \text{pF} + 35\ \text{pF} + [\,1 - (-98.28\,)\,]\ 5\ \text{pF} = 537.4\ \text{pF}$$

$$f_{H_i} = \frac{1}{2\,\pi\,[0.68\ \text{k}\Omega][537.4\ \text{pF}]} = 435.52\ \text{kHz}$$

$$f_{H_o} = \frac{1}{2\,\pi\,R_{Th_o}\,C_o}$$

$$R_{Th_o} = r_o \,\|\, R_C \,\|\, R_L = \infty\ \|\ 4\ \text{k}\Omega\ \|\ 10\ \text{k}\Omega = 2.857\ \text{k}\Omega$$

$$C_o = C_{Wo} + C_{ce} + C_{Mo}$$

$$= 10\ \text{pF} + 1\ \text{pF} + \left[1 - \frac{1}{-98.28}\right] 5\ \text{pF} = 16.05\ \text{pF}$$

$$f_{H_o} = \frac{1}{2\pi\ [2.857\,\text{k}\Omega]\,[16.05\,\text{pF}]} = 3.47\ \text{MHz}$$

(b)

$$f_\beta = \frac{1}{2\,\pi\,\beta_{\text{mid}}\,r_e\,[C_\pi + C_u\,]}$$

$$\beta_{\text{mid}} = \beta = 100$$

$$f_\beta = \frac{1}{2\,\pi\,(100)[29.1\ \Omega][35\ \text{pF} + 5\ \text{pF}]} = 1.367\ \text{MHz}$$

$$f_T = \beta_{\text{mid}}\,f_\beta = (100)\,(1.367\ \text{MHz}\,) = 136.7\ \text{MHz}$$

Example 4.14

For the circuit of Example 4.8 with the same parameter values, calculate the following

(a) f_{H_i} and f_{H_o}
(b) f_β and f_T

Take $C_{Wi} = 7$ pF $C_{Wo} = 11$ pF $C_{bc} = 5$ pF $C_{be} = 20$ pF $C_{ce} = 10$ pF.

Solution

$$\left.\begin{aligned} r_e &= 8.02\ \Omega \\ A_V &= -\ 246.88 \end{aligned}\right\} \quad \text{From example 4.8}$$

(a)

$$f_{H_i} = \frac{1}{2\pi R_{Th_i} C_i}$$

$$R_{Th_i} = R_S \parallel R_B \parallel R_i'$$

$$R_i' = \beta\, r_e = (100)\,(8.02\ \Omega) = 802\ \Omega$$

$$R_{Th_i} = 0.6\ \text{k}\Omega \parallel 500\ \text{k}\Omega \parallel 802\ \Omega = 342.98\ \Omega$$

$$\begin{aligned} C_i &= C_{Wi} + C_{be} + C_{Mi} \\ &= 7\ \text{pF} + 20\ \text{pF} + [1 - (-\ 246.88\ )]\ 5\ \text{pF} \\ &= 1.266\ \text{nF} \end{aligned}$$

$$f_{H_i} = \frac{1}{2\pi[342.98\ \Omega][1.266\ \text{nF}]} = 366.53\ \text{kHz}$$

$$f_{H_o} = \frac{1}{2\ \pi\, R_{Th_o} C_o}$$

$$\begin{aligned} R_{Th_o} &= r_o \parallel R_C \parallel R_L \\ &= \infty \parallel 3.3\ \text{k}\Omega \parallel 5\ \text{k}\Omega \\ &= 1.987\ \text{k}\Omega \end{aligned}$$

$$\begin{aligned} C_o &= C_{Wo} + C_{ce} + C_{Mo} \\ &= 11\ \text{pF} + 10\ \text{pF} + \left[1 - \frac{1}{-\ 246.88}\right] 5\ \text{pF} \\ &= 26.02\ \text{pF} \end{aligned}$$

$$f_{H_o} = \frac{1}{2\pi[1.987\ \text{k}\Omega]\,[26.02\ \text{pF}]} = 3.07\ \text{MHz}$$

(b)

$$\begin{aligned} f_\beta &= \frac{1}{2\pi\,\beta_{\text{mid}}\ r_e\ [C_{be}+C_{bc}]} \\ &= \frac{1}{2\ \pi\ [100]\,[8.02\ \Omega]\,[20\ \text{pF}+5\ \text{pF}]} \\ &= 7.93\ \text{MHz} \end{aligned}$$

$$\begin{aligned} f_T &= \beta_{\text{mid}}\ f_\beta \\ &= (100)\ (7.93\ \text{MHz}) = 793\ \text{MHz} \end{aligned}$$

Example 4.15

Consider the circuit of Example 4.9 with the same parameter values. Calculate the following

(a) f_{H_i} and f_{H_o}

(b) f_β and f_T

Take $C_{Wi} = 7$ pF $C_{Wo} = 10$ pF $C_{bc} = 15$ pF $C_{be} = 25$ pF $C_{ce} = 12$ pF.

Solution

$$\left.\begin{aligned} r_e &= 23.21\ \Omega \\ Z_b &= 146.6\ \text{k}\Omega \end{aligned}\right\} \text{From example 4.9}$$

(a)

$$f_{H_i} = \frac{1}{2\pi R_{Th_i} C_i}$$

$$R_{Th_i} = R_S \,||\, R_1 \,||\, R_2 \,||\, R_i'$$

But $R_i' = Z_b$ [since configuration is CC]

$$R_{Th_i} = 1\ \text{k}\Omega \,||\, 120\ \text{k}\Omega \,||\, 33\ \text{k}\Omega \,||\, 146.6\ \text{k}\Omega$$

$$= 956.5\ \Omega$$

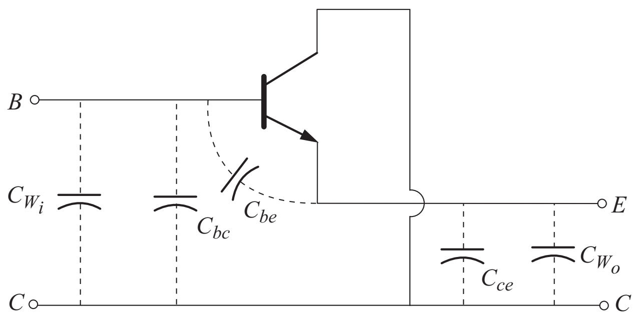


Fig. A

From Fig(A)

$$C_i = C_{Wi} + C_{bc} + C_{be}$$

$$= 7\ \text{pF} + 15\ \text{pF} + 25\ \text{pF}$$

$$= 47\ \text{pF}$$

$$f_{H_i} = \frac{1}{2\pi[956.5\ \Omega][47\ \text{pF}]}$$

$$= 3.54\ \text{MHz}$$

$$f_{H_o} = \frac{1}{2\pi R_{Th_o} C_o}$$

$$R_{Th_o} = R_E \parallel R_L \parallel r_e$$
$$= 2\ \text{k}\Omega \parallel 5\ \text{k}\Omega \parallel 23.21\Omega$$
$$= 22.83\ \Omega$$

From Fig A

$$C_o = C_{Wo} + C_{ce}$$
$$= 10\ \text{pF} + 12\ \text{pF} = 22\ \text{pF}$$
$$f_{H_o} = \frac{1}{2\pi(22.83\ \Omega)(22\ \text{pF})} = 316.87\ \text{MHz}$$

(b)

$$f_\beta = \frac{1}{2\pi\beta_{\text{mid}}\, r_e\, [C_{be}+C_{bc}]}$$
$$= \frac{1}{2\pi(100)(23.21\ \Omega)[25\ \text{pF}+15\ \text{pF}]}$$
$$= 1.714\ \text{MHz}$$
$$f_T = \beta_{\text{mid}}\, f_\beta = (100)(1.714\ \text{MHz}) = 171.4\ \text{MHz}$$

Example 4.16

For the circuit of Example 4.10 with the same parameter values calculate the following

(a) f_{H_i} and f_{H_o}

(b) f_β and f_T

Take $C_{Wi} = 10$ pF $C_{Wo} = 10$ pF $C_{bc} = 18$ pF $C_{be} = 24$ pF $C_{ce} = 12$ pF.

Solution

$$r_e = 9.09\ \Omega \qquad \text{(From example 4.10)}$$

(a)

$$f_{H_i} = \frac{1}{2\pi R_{Th_i} C_i}$$
$$R_{Th_i} = R_S \parallel R_E \parallel R_i'$$
$$R_i' = r_e \qquad \text{[since configuration is CB]}$$
$$R_{Th_i} = 120\ \Omega \parallel 1.5\ \text{k}\Omega \parallel 9.09\ \Omega = 8.4\ \Omega$$

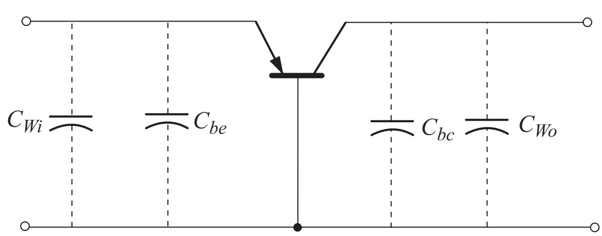


Fig. A

From Fig. A

$$C_i = C_{W_i} + C_{be} = 10 \text{ pF} + 24 \text{ pF} = 34 \text{ pF}$$

$$f_{H_i} = \frac{1}{2\pi(8.4\ \Omega)(34 \text{ pF})} = 557.26 \text{ MHz}$$

$$f_{H_o} = \frac{1}{2\pi R_{Th_o} C_o}$$

$$R_{Th_o} = r_o \parallel R_C \parallel R_L$$

$$= \infty \parallel 3 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega$$

$$= 1.83 \text{ k}\Omega$$

$$C_o = C_{bc} + C_{W_o} \qquad \text{[see Fig A]}$$

$$= 18 \text{ pF} + 10 \text{ pF} = 28 \text{ pF}$$

$$f_{H_o} = \frac{1}{2\pi (1.83 \text{ k}\Omega)(28 \text{ pF})} = 3.1 \text{ MHz}$$

(b)

$$f_\beta = \frac{1}{2\pi \beta_{\text{mid}} r_e [C_{be} + C_{bc}]}$$

$$= \frac{1}{2\pi [75][9.09\ \Omega][24 \text{ pF} + 18 \text{ pF}]}$$

$$= 5.55 \text{ MHz}$$

$$f_T = \beta_{\text{mid}} f_\beta = (75)(5.55 \text{ MHz})$$

$$= 416.25 \text{ MHz}$$

Exercise Problems

4.1 The following data is available for an amplifier

$P_i = 100$ W $\quad P_o = 50$ W $\quad V_i = 100$ V $\quad R_o = 20\ \Omega$

Calculate (a) Power gain in *dB* (b) Voltage gain in *dB*.

Hint: $A_{P\,(dB)} = 10 \log_{10} [P_o / P_i]$ $\qquad A_{V\,(dB)} = 20 \log_{10} [V_o / V_i]$

$$P_o = \frac{V_o^2}{R_L} = \frac{V_o^2}{R_o} \qquad P_i = \frac{V_i^2}{R_i}$$

4.2 Two voltage measurements made across the same resistance are $V_1 = 20$ V and $V_2 = 100$ V. Calculate

(a) Power gain in dB of the second reading over the first reading

(b) Voltage gain

Hint: $R_i = R_o$

4.3 If the applied ac power to a system is 5 μW at 100 mV and the output power is 48 W, calculate

(a) Power gain in dB (b) Voltage gain in dB if $R_L = 40\ \text{k}\Omega$

(c) Input impedance (d) Output impedance

4.4 An amplifier is required to deliver 50 W to a 16Ω loud speaker. Calculate

(a) The input power required if the power gain is 20 dB

(b) The input Voltage required if the amplifier voltage gain is 40 dB.

4.5 The total decibel Voltage gain of a three stage system is 120 dB. The second stage has twice the decibel voltage gain of the first and the third has 2.7 times the decibel gain of the first. Calculate

(a) The decibel voltage gain of each stage

(b) The voltage gain of each stage and the overall voltage gain

Hint: $A_V = A_{V_1} \cdot A_{V_2} \cdot A_{V_2}$

$$A_{V\,(dB)} = A_{V_1\,(dB)} + A_{V_2\,(dB)} + A_{V_3\,(dB)}$$

Chapter 5

GENERAL AMPLIFIERS

When the amplification from a single stage amplifier is not sufficient for a particular purpose, or when the input or output impedance is not of suitable magnitude for the intended application, two or more amplifier stages are connected in cascade. The cascade of CE and CB stages is called cascode amplifier. This chapter discusses cascaded stages, cascode amplifier, current mirror and current sources. Analysis of transistor configurations using the approximate and the complete hybrid model has also been considered.

5.1 CASCADED SYSTEMS

When the amplification from a single stage amplifier is not sufficient for a particular purpose or when the input or output impedance is not of suitable magnitude for the intended application, two or more amplifier stages may be connected in cascade: i.e., the output of a given stage is connected to the input of the next stage. Such an arrangement is called multistage amplifier. Figure 5.1 shows three amplifier stages connected in cascade. The two port system approach is very much useful in the analysis of cascaded systems.

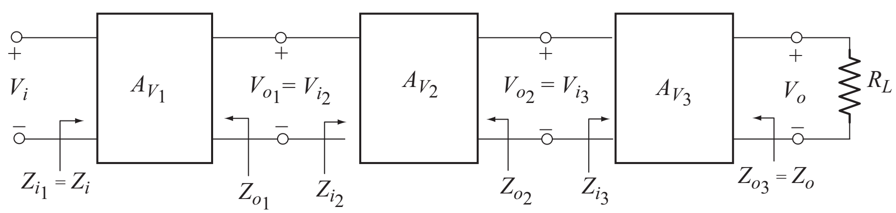


Fig. 5.1 *Cascaded amplifiers*

It is important to note that, the input impedance of a given stage loads the output of the preceeding stage. Thus the voltage gains A_{V_1}, A_{V_2} and A_{V_3} are the loaded voltage gains whose values can be calculated using the corresponding no load gains. Also for the cascaded system, the input impedance is that of the first stage and the output impedance is that of the last stage.

i.e., $$Z_i = Z_{i_1} \quad \text{and} \quad Z_o = Z_{o_3}$$

Since the output of one stage is connected as input to the next stage,

$$V_{i_2} = V_{o_1} \text{ and } V_{i_3} = V_{o_2}$$

The total or the overall voltage gain of the cascaded system can be obtained as follows:

$$A_{V_T} = \frac{V_o}{V_i}$$

$$A_{V_T} = \frac{V_o}{V_{i_3}} \cdot \frac{V_{i_3}}{V_{i_2}} \cdot \frac{V_{i_2}}{V_i}$$

Using $V_{i_3} = V_{o_2}$ and $V_{i_2} = V_{o_1}$ we have

$$A_{V_T} = \frac{V_o}{V_{i_3}} \cdot \frac{V_{o_2}}{V_{i_2}} \cdot \frac{V_{o_1}}{V_i}$$

$$= A_{V_3} \cdot A_{V_2} \cdot A_{V_1}$$

or

$$A_{V_T} = A_{V_1} \cdot A_{V_2} \cdot A_{V_3} \tag{5.1}$$

where

$$A_{V_1} = \frac{V_{o_1}}{V_{i_1}} = \text{Loaded voltage gain of stage 1}$$

$$A_{V_2} = \frac{V_{o_2}}{V_{i_2}} = \text{Loaded voltage gain of stage 2}$$

$$A_{V_3} = \frac{V_{o_3}}{V_{i_3}} = \text{Loaded voltage gain of stage 3}$$

For n cascaded amplifier stages, the total voltage gain is given by

$$A_{V_T} = A_{V_1} \cdot A_{V_2} \cdot A_{V_3} \cdots A_{V_n} \tag{5.2}$$

The total current gain is given by

$$A_{I_T} = -A_{V_T} \frac{Z_{i_1}}{R_L} \tag{5.3}$$

Example 5.1

For the cascaded arrangement shown below, determine

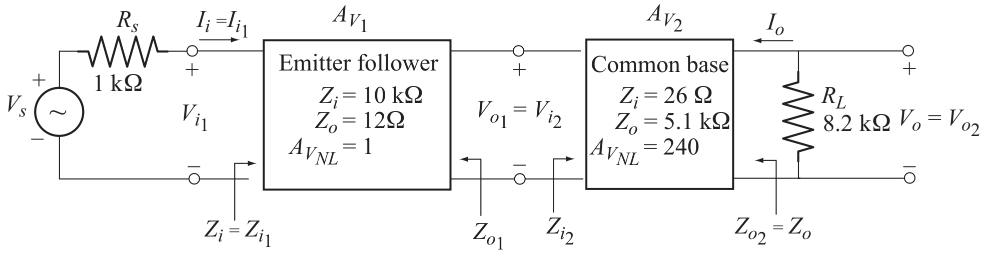
(a) The loaded gain for each stage.

(b) The total gain for the system, A_V and A_{V_S}.

(c) The total current gain for the system.

(d) The total gain for the system if the emitter-follower configuration were removed.

(e) The phase relation ship between V_o and V_i.



Solution

(a) For the emitter-follower, the load is Z_{i_2},

$$A_{V_1} = \frac{V_{0_1}}{V_{i_1}} = \frac{Z_{i_2}}{Z_{i_2} + Z_{o_1}} A_{V_{NL}} = \frac{26\Omega}{26\Omega + 12\Omega} = 0.684$$

For the common-base configuration,

$$A_{V_2} = \frac{V_o}{V_{i_2}} = \frac{R_L}{R_L + Z_{o_2}} A_{V_{NL}}$$

$$= \frac{8.2\,\text{k}\Omega}{8.2\,\text{k}\Omega + 5.1\,\text{k}\Omega}(240) = 147.97$$

(b) The total voltage gain is

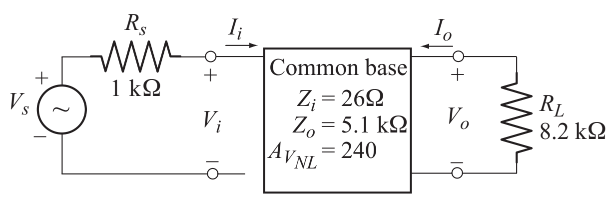
$$A_{V_T} = A_{V_1} A_{V_2} = (0.684)\,(147.97) = 101.20$$

$$A_{V_S} = \frac{Z_{i_1}}{Z_{i_1} + R_S} A_{V_T} = \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 1\,\text{k}\Omega}(101.20) = 92$$

(c)

$$A_{I_T} = -A_{V_T}\frac{Z_{i_1}}{R_L} = -(101.\,20)\frac{10\,\text{k}\Omega}{8.2\,\text{k}\Omega} = -123.\,41$$

(d) With out emitter-follower, the arrangement is as shown below.



$$A_V = \frac{V_o}{V_i} = \frac{R_L}{Z_o + R_L} A_{V_{NL}}$$

$$= \frac{8.2\,\text{k}\Omega}{5.1\,\text{k}\Omega + 8.2\,\text{k}\Omega}(240)$$

$$= 147.97$$

$$A_{V_S} = \frac{V_o}{V_s} = \frac{V_o}{V_i}\frac{V_i}{V_s}$$

$$= A_V \frac{Z_i}{R_s + Z_i} = (147.97)\frac{26\Omega}{1\text{k}\Omega + 26\Omega} = 3.75$$

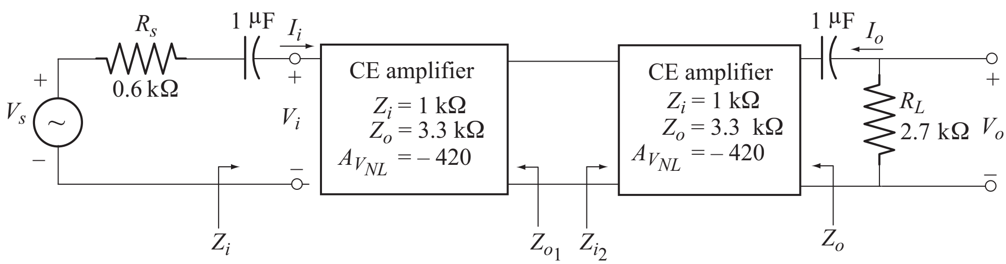
(e) Both emitter-follower and common-base stage introduces zero degree phase shift. Hence V_o and V_i are in phase.

Note : When the emitter-follower is removed, the voltage gain falls from 92 to 3.75. This is due to the very low input impedance of CB stage. Emitter-follower matches the low input impedance of CB stage with the high source resistance R_S.

Example 5.2

For the cascaded arrangement shown below calculate

(a) The loaded voltage gain of each stage
(b) The total gain of the system, A_V and A_{Vs}
(c) The loaded current gain of each stage
(d) The total current gain of the system
(e) How Z_i is affected by the second stage and R_L
(f) How Z_o is affected by the first stage and R_s
(g) The phase relation ship between V_o and V_i



Solution

(a) The load on first stage is Z_{i_2}

$$A_{V_1} = \frac{Z_{i_2}}{Z_{o_1} + Z_{i_2}} A_{V_{NL}} = \frac{1\text{k}\Omega}{3.3\text{k}\Omega + 1\text{k}\Omega}(-420) = -97.67$$

$$A_{V_2} = \frac{R_L}{Z_o + R_L} A_{V_{NL}} = \frac{2.7\text{k}\Omega}{3.3\text{k}\Omega + 2.7\text{k}\Omega}(-420) = -189$$

(b) The total voltage gain is

$$A_{V_T} = A_{V_1} A_{V_2} = (-97.67)(-189) = 18.45 \times 10^3$$

$$A_{V_s} = \frac{Z_i}{R_s + Z_i} A_{V_T}$$

$$= \frac{1\text{k}\Omega}{0.6\text{k}\Omega + 1\text{k}\Omega}(18.45 \times 10^3) = 11.53 \times 10^3$$

(c)
$$A_{I_1} = -A_{V_1}\frac{Z_i}{Z_{i_2}} \quad \text{[Since load on first stage is } Z_{i_2}\text{]}$$

$$= -(-97.67)\frac{1\text{k}\Omega}{1\text{k}\Omega} = 97.67$$

$$A_{I_2} = -A_{V_2}\frac{Z_{i_2}}{R_L} = -(-189)\frac{1\text{k}\Omega}{2.7\text{k}\Omega} = 70$$

(d) Total current gain is

$$A_{I_T} = A_{I_1} \cdot A_{I_2} = (97.67)(70) = 6.84 \times 10^3$$

(e) Z_i is unaffected by the second stage and R_L

(f) Z_o is unaffected by the first stage and R_S

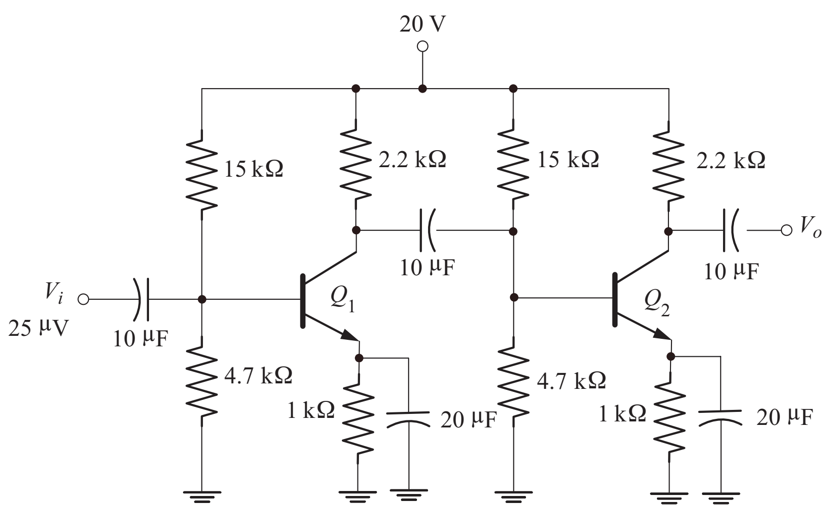
(g) Each CE stage introduces a phase shift of 180º. Hence the phase shift between V_o and V_i is 360° or 0º. i.e., V_o and V_i are in phase.

Example 5.3

For the BJT cascade amplifier shown below

(a) Calculate the dc bias voltages and collector current for each stage.
(b) Calculate the voltage gain of each stage, the overall voltage gain and the output voltage.
(c) Repeat part (b) with a 10 kΩ load applied to the second stage.
(d) Calculate the input impedance of the first stage and the output impedance of the second stage.

Take $\beta = 200$ for both transistors.



Solution

(a) Since R_1, R_2, R_E, R_C and β are same for both the transistors, the dc bias voltages and currents are identical in both the stages.

Check for $\beta R_E \geq 10\ R_2$

$$\beta R_E = (200)\ (1\ \text{k}\Omega) = 200\ \text{k}\Omega$$
$$10\ R_2 = (10)\ (4.7\ \text{k}\Omega) = 47\ \text{k}\Omega$$

Since $\beta R_E > 10\ R_2$, We can use approximate analysis.

$$V_B = \frac{V_{CC}\,R_2}{R_1 + R_2} = \frac{(20\,\text{V})(4.7\,\text{k}\Omega)}{15\,\text{k}\Omega + 4.7\,\text{k}\Omega} = 4.77\ \text{V}$$
$$V_E = V_B - V_{BE} = 4.77\ \text{V} - 0.7\ \text{V} = 4.07\ \text{V}$$
$$I_E = \frac{V_E}{R_E} = \frac{4.07\,\text{V}}{1\text{K}\Omega} = 4.07\ \text{mA}$$
$$r_e = \frac{26\,\text{mV}}{I_E} = \frac{26\,\text{mV}}{4.07\,\text{mA}} = 6.39\ \Omega$$
$$I_B = \frac{I_E}{1+\beta} = \frac{4.07\,\text{mA}}{201} = 20.25\ \mu\text{A}$$
$$I_C = \beta I_B = (200)\ (20.25\ \mu\text{A}) = 4.05\ \text{mA}$$
$$V_C = V_{CC} - I_C R_C$$
$$= 20\ \text{V} - (4.05\ \text{mA})\ (2.2\ \text{k}\Omega) = 11.09\ \text{V}$$
$$V_{CE} = V_C - V_E = 11.09\ \text{V} - 4.07\ \text{V} = 7.02\ \text{V}$$

(b) Load on the first stage is

$$R_{L_1} = R_C \parallel Z_{i_2}$$
$$Z_{i_2} = R_1 \parallel R_2 \parallel \beta\, r_e$$
$$= 15\ \text{k}\Omega \parallel 4.7\ \text{k}\Omega \parallel (200)\ (6.39\ \Omega) = 0.942\ \text{k}\Omega$$
$$R_{L_1} = 2.2\ \text{k}\Omega \parallel 0.942\ \text{k}\Omega = 0.659\ \text{k}\Omega$$
$$A_{V_1} = -\frac{R_{L_1}}{r_e} = -\frac{0.659\,\text{k}\Omega}{6.39\,\Omega} = -103.13$$

Since second stage is unloaded, its voltage gain is

$$A_{V_2(NL)} = -\frac{R_C}{r_e} = -\frac{2.2\,\text{k}\Omega}{6.39\,\Omega} = -344.28$$

Overall Voltage gain is

$$A_{V_T(NL)} = A_{V_1}\,A_{V_2(NL)} = (-103.13)\ (-344.28) = 35.5 \times 10^3$$
$$V_o = A_{V_T(NL)}\,V_i = (35.5 \times 10^3)\ (25\ \mu\text{V}) = 887.5\ \text{mV}$$

(c) The overall voltage gain with $R_L = 10$ kΩ is

$$A_{V_T} = \frac{V_o}{V_i} = \frac{R_L}{R_L + Z_o} A_{V_T(NL)}$$

But $$Z_o = R_C = 2.2 \text{ k}\Omega$$

$$A_{V_T} = \frac{10\text{k}\Omega}{10\text{k}\Omega + 2.2\text{k}\Omega}(35.5 \times 10^3) = 29.09 \times 10^3$$

$$V_o = A_{V_T} V_i = (29.09 \times 10^3)\,(25\ \mu\text{V}) = 727.\ 25 \text{ mV}$$

(d) Input impedance of the first stage is

$$Z_{i_1} = Z_i = R_1 \parallel R_2 \parallel \beta r_e = 0.942 \text{ k}\Omega \qquad \text{(same as } Z_{i_2}\text{)}$$

Output impedance of second stage is

$$Z_{o_2} = Z_o = R_C = 2.2 \text{ k}\Omega$$

Note: If the two stages are not identical then dc analysis has to be carried out separately and their r_e values are different.

5.2 CASCODE CONNECTION

In cascode connection the output of CE stage drives the input of CB stage as shown in Fig. 5.2. The cascode connection has low input capacitance, which is an advantage at high frequencies such as VHF and UHF. At these higher frequencies, the input capacitance becomes a limiting factor on the voltage gain. With a cascode amplifier, the low input capacitance allows the circuit to amplify higher frequencies than are possible with only a CE amplifier.

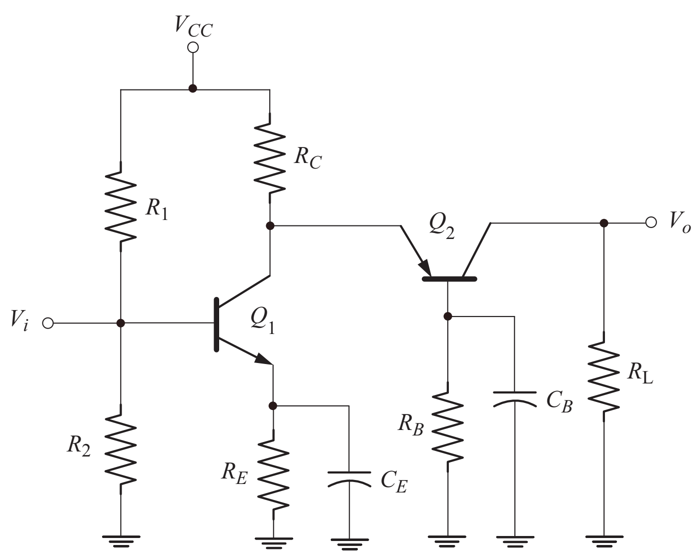


Fig. 5.2 Cascode configuration

Cascode connection has high input impedance (provided by CE) and high output impedance (provided by CB). The CB stage provides an excellent high frequency response.

Due to direct coupling between CE and CB stages, the dc bias current for Q_2 is obtained from the collector node of Q_2. R_B is used to limit the bias current of Q_2. For ac operation, R_B is shorted out by C_B. Thus the base of Q_2 will be at ground potential for ac operation, which is essential for CB configuration.

The low input impedance of CB stage loads the output of CE stage. Thus the voltage gain of CE stage is very low. A large voltage gain is provided by the CB stage, thus the overall voltage gain is high with a good input impedance level.

An alternate cascode configuration is shown in Fig .5.3. In this circuit Q_1 is in CE configuration and Q_2 in CB configuration. For ac operation, C_2 shorts the base of Q_2 to ground. Note that both Q_1 and Q_2 are npn transistors.

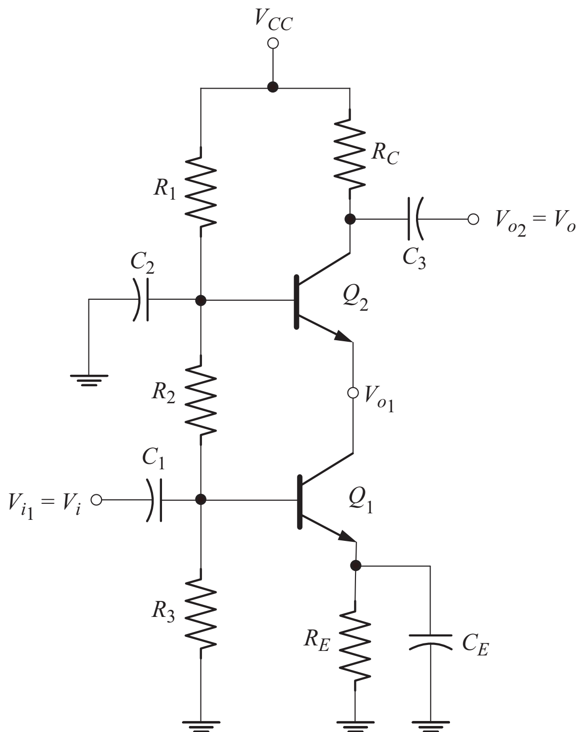


Fig. 5.3 Alternate cascode connection

Example 5.4

For the cascode connection of Fig. 5.3 the followings data are available:

$R_1 = 6.8\ \text{k}\Omega$ $\quad R_2 = 5.6\ \text{k}\Omega$ $\quad R_3 = 4.7\ \text{k}\Omega$

$R_C = 1.8\ \text{k}\Omega$ $\quad R_E = 1.1\ \text{k}\Omega$ $\quad V_{CC} = 18\ \text{V}$

$\beta_1 = \beta_2 = 200$ $\quad V_i = 10\ \text{mV}$

(a) Calculate the dc bias voltages V_{B_1}, V_{B_2}, V_{C_2}

(b) Calculate the no load voltage gain and the output voltage $V_{o_2} = V_o$

(c) Calculate the voltage gain with a load of 10 kΩ connected to the second stage and the output voltage V_o

(d) Input and output impedances

Solution

(a) DC voltage at the base of Q_1 is

$$V_{B_1} = \frac{R_3}{R_1 + R_2 + R_3} V_{CC}$$

$$= \frac{4.7\,\text{k}\Omega}{6.8\,\text{k}\Omega + 5.6\,\text{k}\Omega + 4.7\,\text{k}\Omega} (18\text{ V}) = 4.9\text{ V}$$

$$V_{E_1} = V_{B_1} - V_{BE} = 4.9\text{ V} - 0.7\text{ V} = 4.2\text{ V}$$

$$I_{E_1} \approx I_{C_1} = \frac{V_{E_1}}{1.1\,\text{k}\Omega} = \frac{4.2\text{ V}}{1.1\,\text{k}\Omega} = 3.82\text{ mA}$$

Note that the emitter current of Q_2 is also the collector current of Q_1

$$\therefore \qquad I_{E_2} = I_{C_1} = 3.82\text{ mA}$$

Since $I_{E_1} = I_{E_2}$, the dynamic resistance r_e is the same for both the transistor.

$$r_e = \frac{26\text{ mV}}{3.82\text{ mA}} = 6.8\ \Omega$$

Voltage on the base of Q_2 is

$$V_{B_2} = \frac{R_3 + R_2}{R_1 + R_2 + R_3} V_{CC}$$

$$= \frac{4.7\,\text{k}\Omega + 5.6\,\text{k}\Omega}{6.8\,\text{k}\Omega + 5.6\,\text{k}\Omega + 4.7\,\text{k}\Omega} (18\text{ V}) = 10.84\text{ V}$$

$$I_{C_2} = \frac{V_{CC} - V_{C_2}}{R_C}$$

$$\Rightarrow \qquad V_{C_2} = V_{CC} - I_{C_2} R_C = 18\text{ V} - (3.82\text{ mA})(1.8\text{ k}\Omega) = 11.124\text{ V}$$

(b) The load on the transistor Q_1 is the input impedance of the Q_2. Since Q_2 is in the CB configuration, $R_C = r_e$ for Q_1.

$$A_{V_1} = -\frac{R_C}{r_e} = -\frac{r_e}{r_e} = -1$$

For the second stage (CB)

$$A_{V_2} = \frac{R_C}{r_e} = \frac{1.8\,\text{k}\Omega}{6.8\,\Omega} = 264.7$$